Nonclassicality, quasiprobabilities and weak values explored in neutron interferometry

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Disclaimer

Weak values, weak measurements and weak interactions are not related to the weak force!

Overview

- Introduction
- Nonclassicality

5 min break for questions

- Quasiprobabilities and weak values
- Experimental results

Introduction

Motivation and goal

Quantum mechanics exhibits features with no classical counterpart.

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However, not all quantum mechanics is non-classical and identifying the classical/quantum boundary is of importance.

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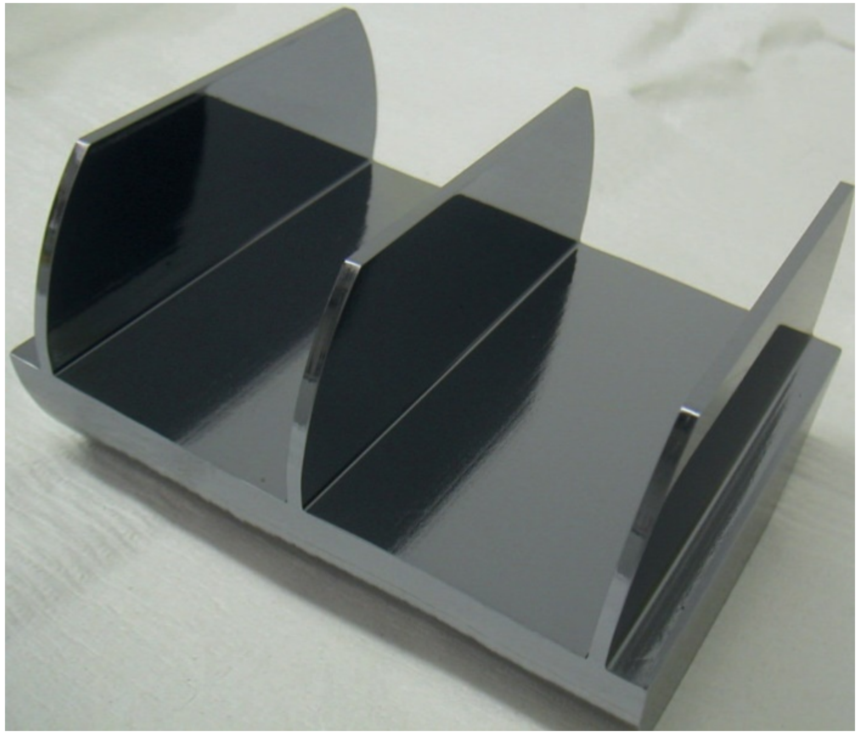
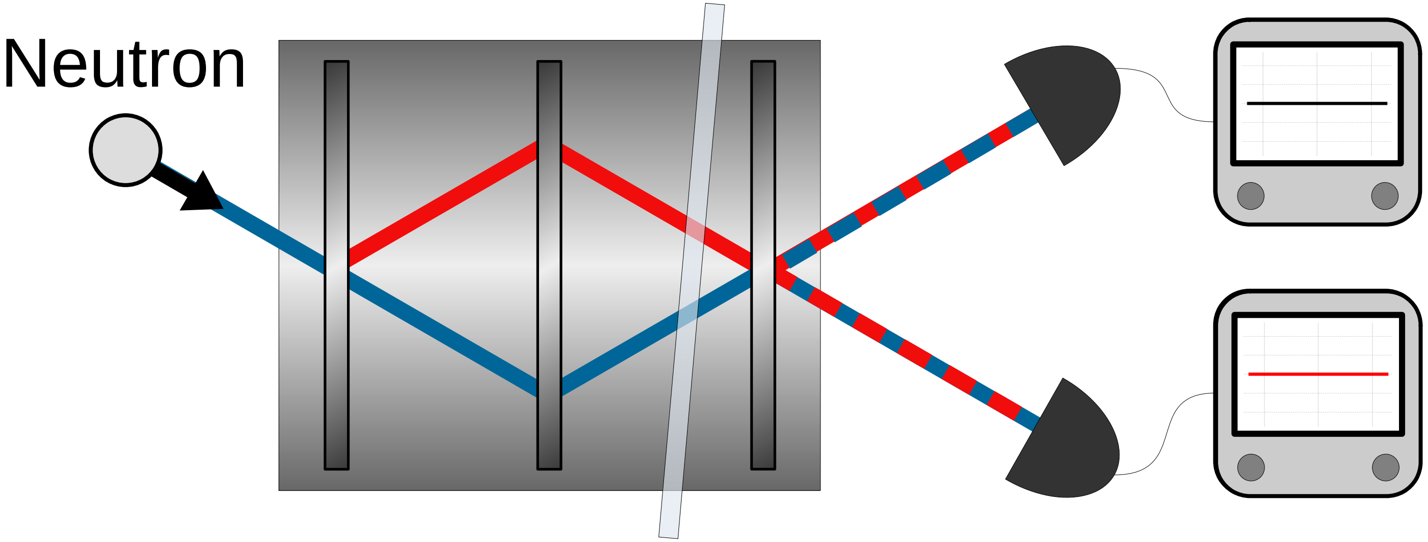
However, not all quantum mechanics is non-classical and identifying the classical/quantum boundary is of importance.

We are interested in experimental tools to probe such boundaries.

Neutron Mach-Zehnder interferometer:





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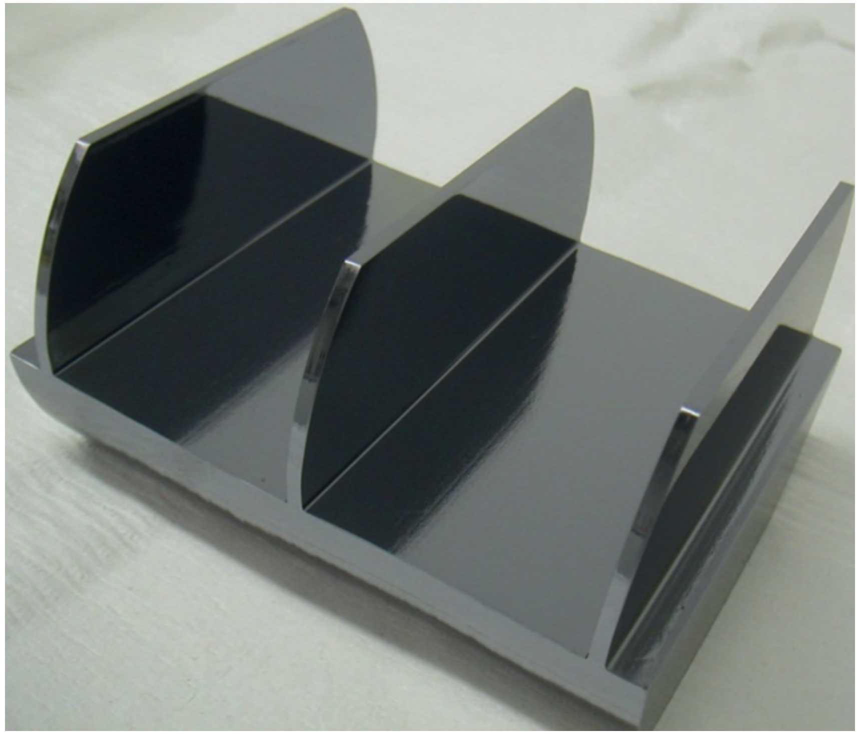

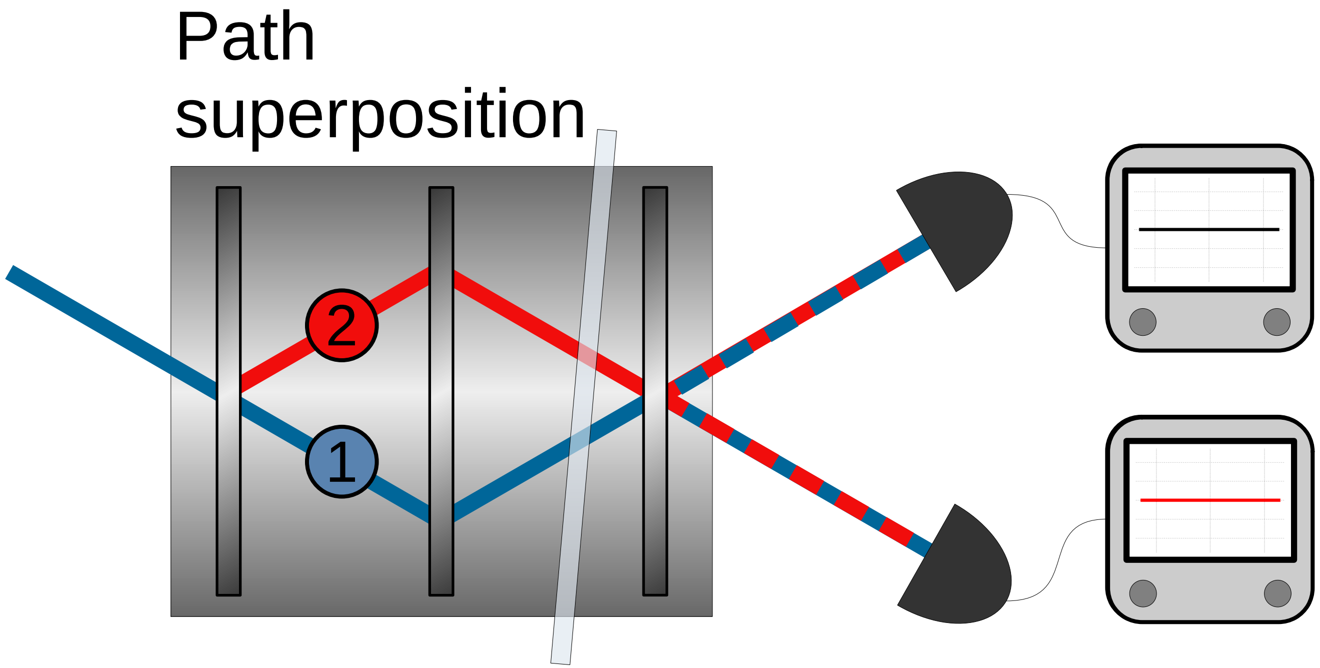

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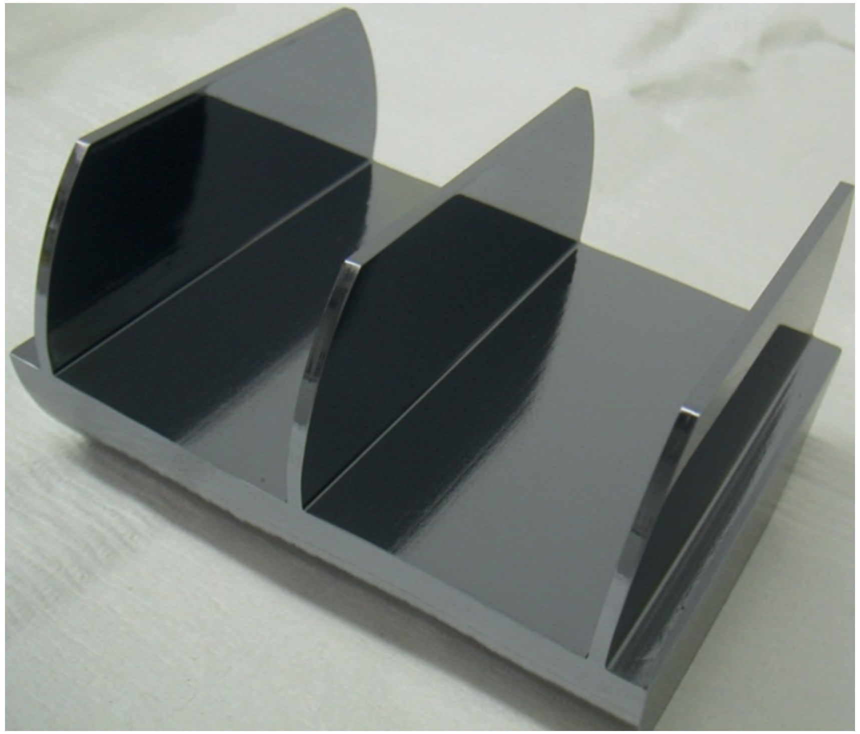

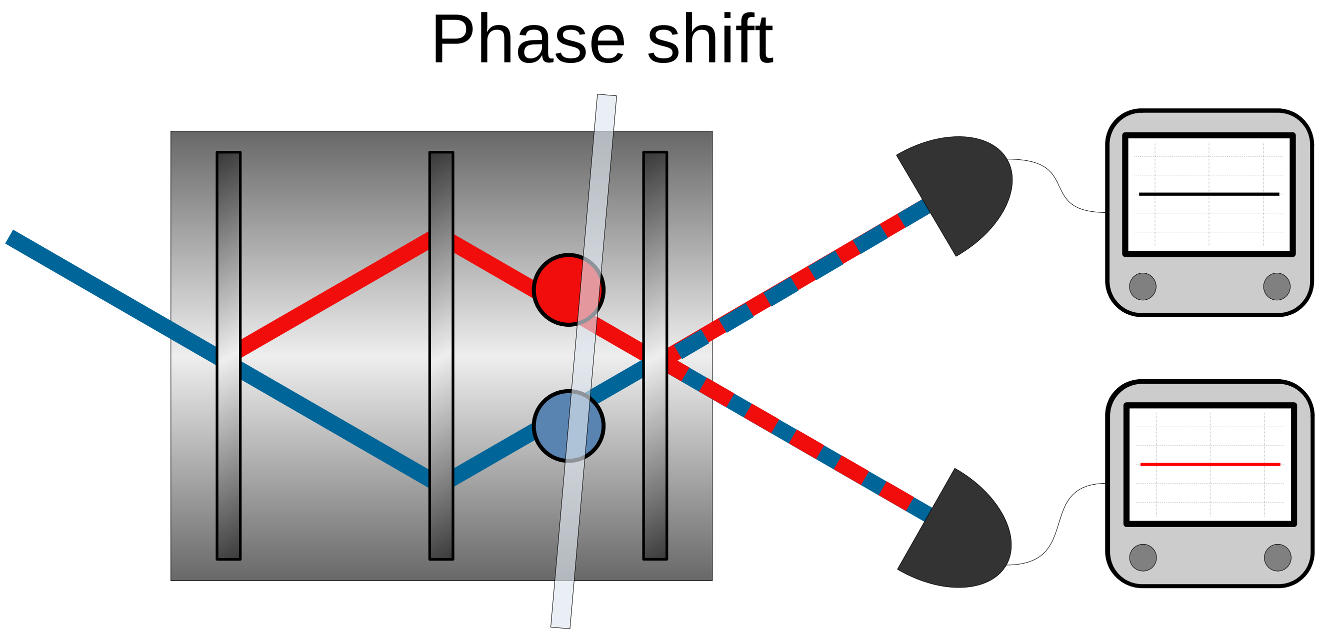

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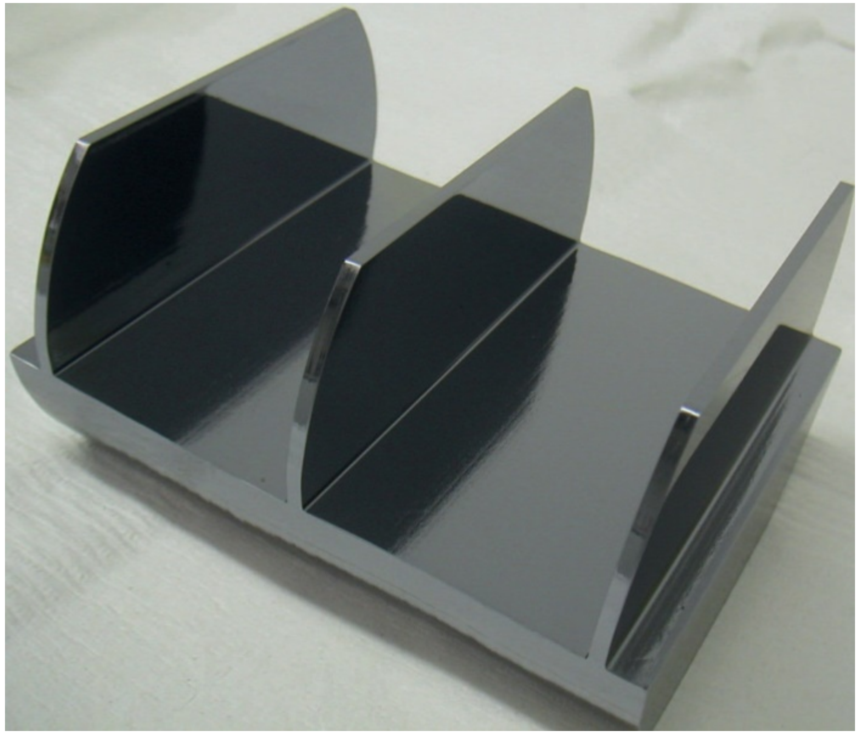

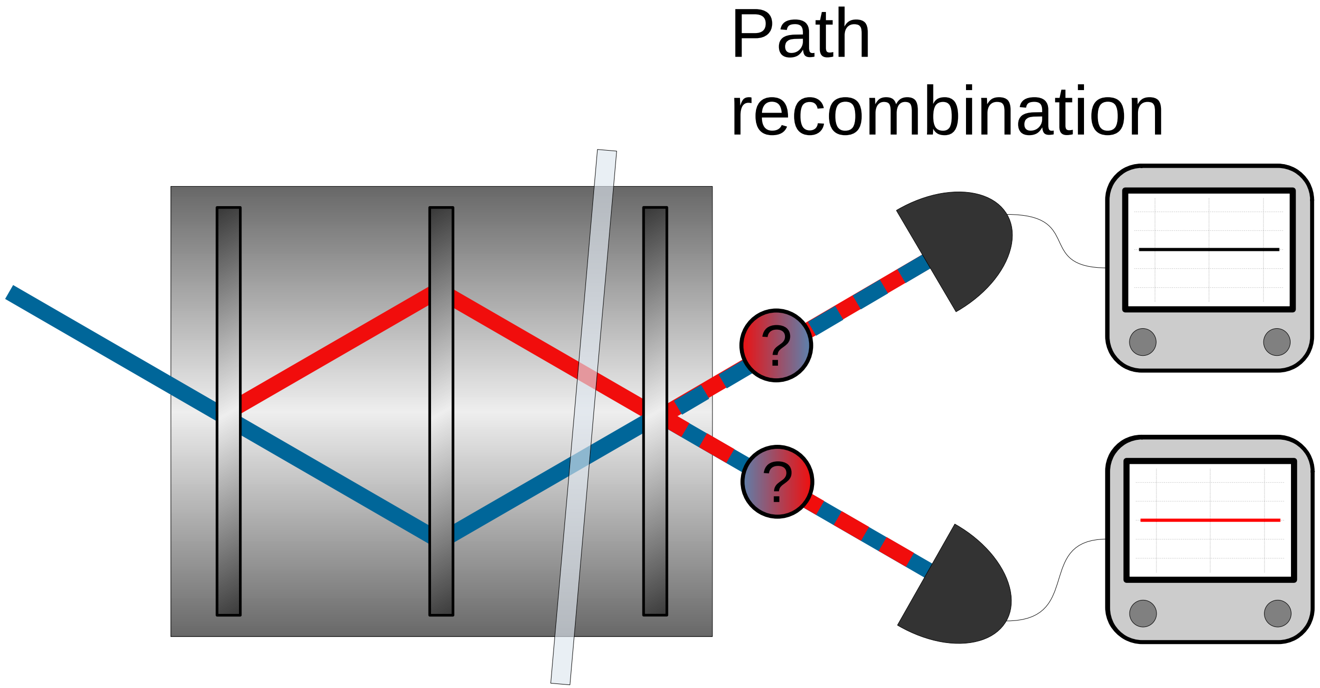

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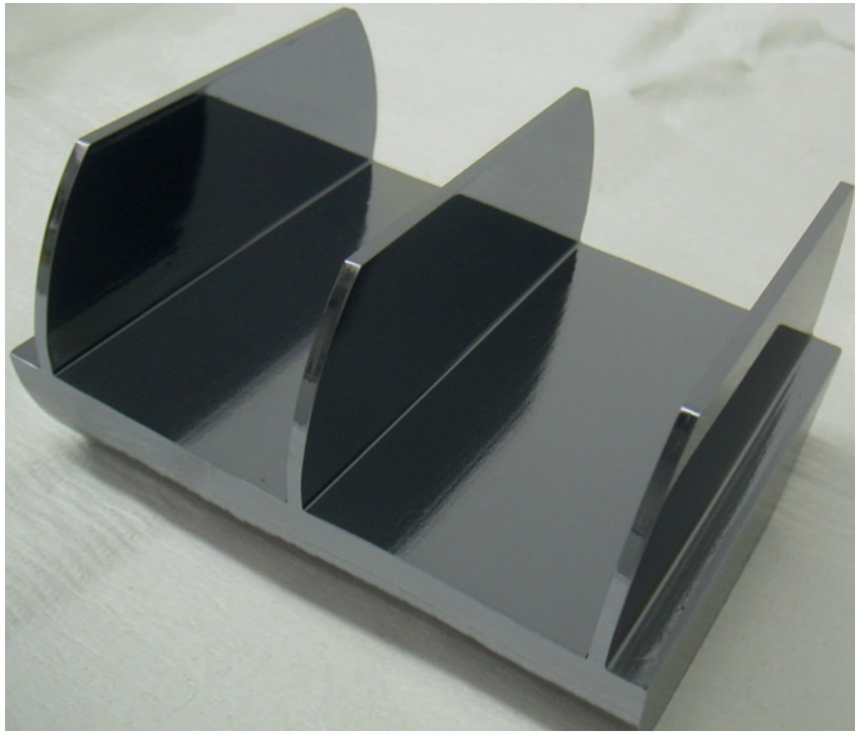
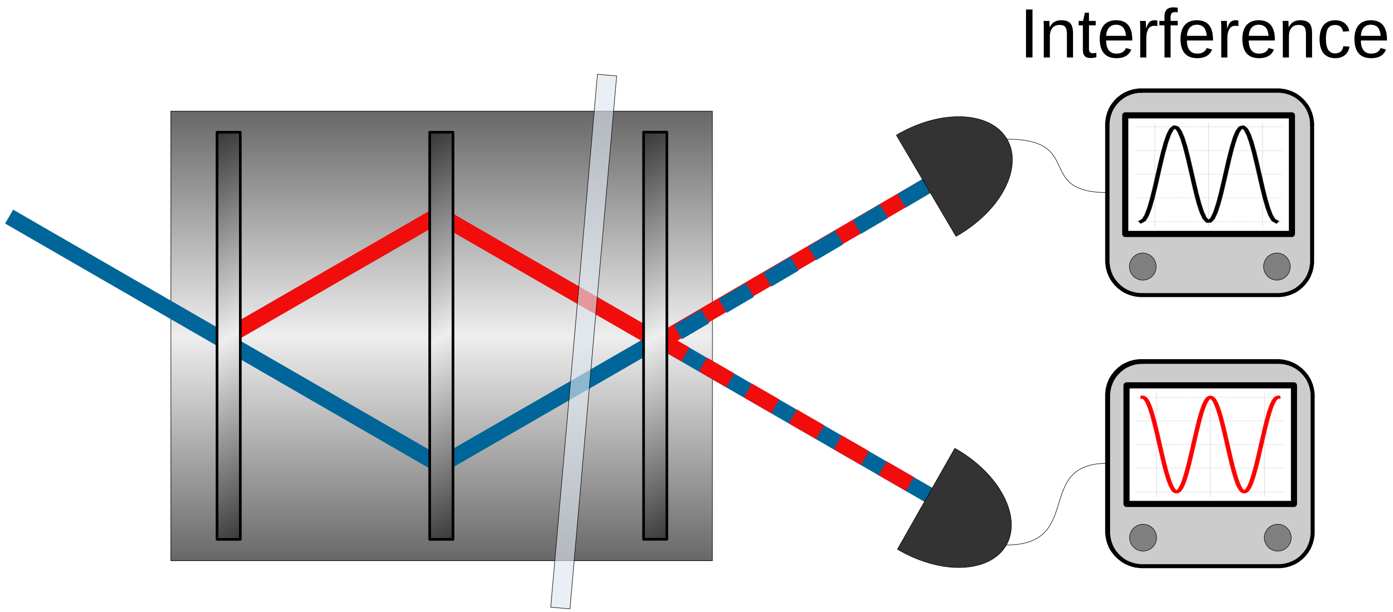




Neutron interferometry

Why neutron interferometry?

- Interference is intrinsically a single-massive-particle phenomenon.
- Matter-wave interferometry at room temperature and ambient pressure, with macroscopic beam separation and high contrast.
- Manipulation of several degrees of freedom (path, spin, energy).

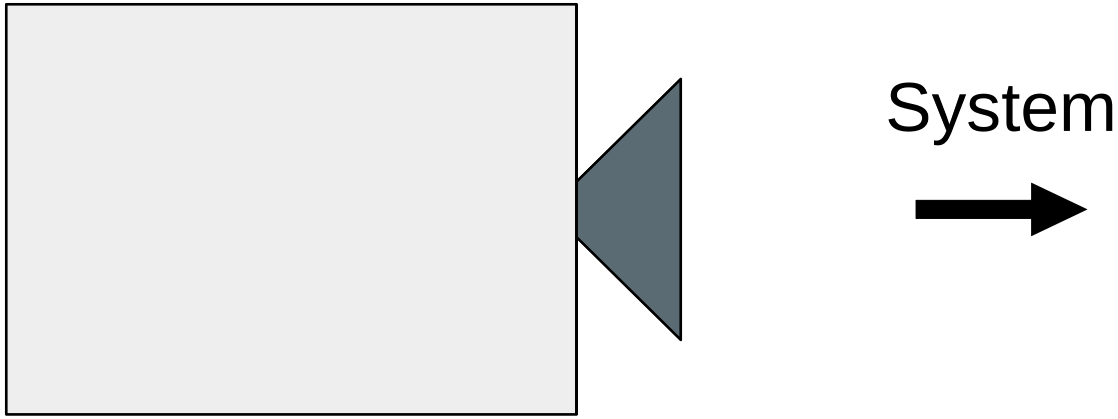
Sponar, S., Sedmik, R. I., Pitschmann, M., Abele, H. & Hasegawa, Y. Tests of fundamental quantum mechanics and dark interactions with low-energy neutrons. Nat. Rev. Phys. 3, 309–327 (2021).

Nonclassicality

Preparation and Measurement

Preparation (P)

Preparation (P)



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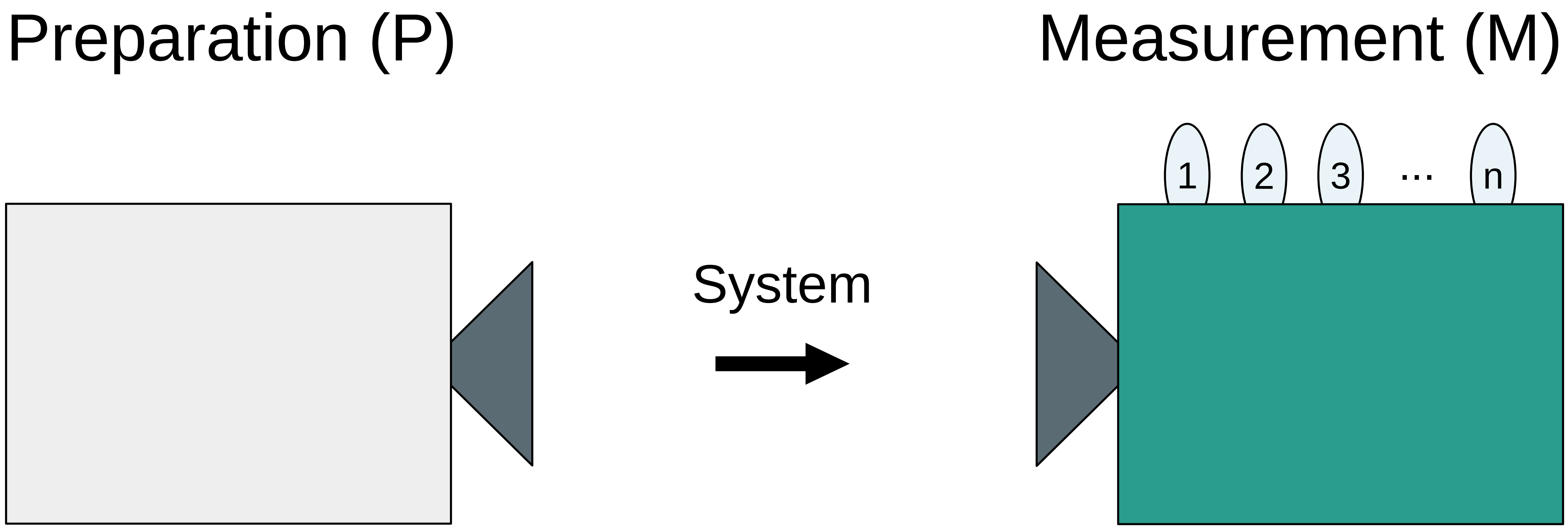
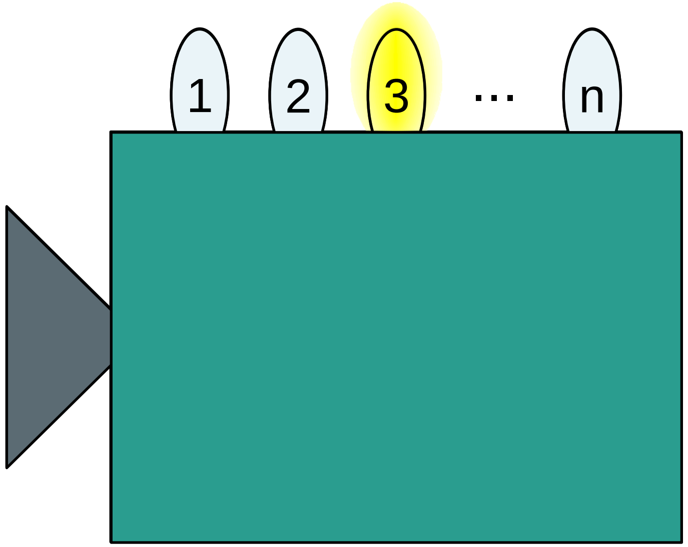


Preparation and Measurement

Preparation (P)

Measurement (M)



Preparation and Measurement

Preparation (P)

Measurement (M)

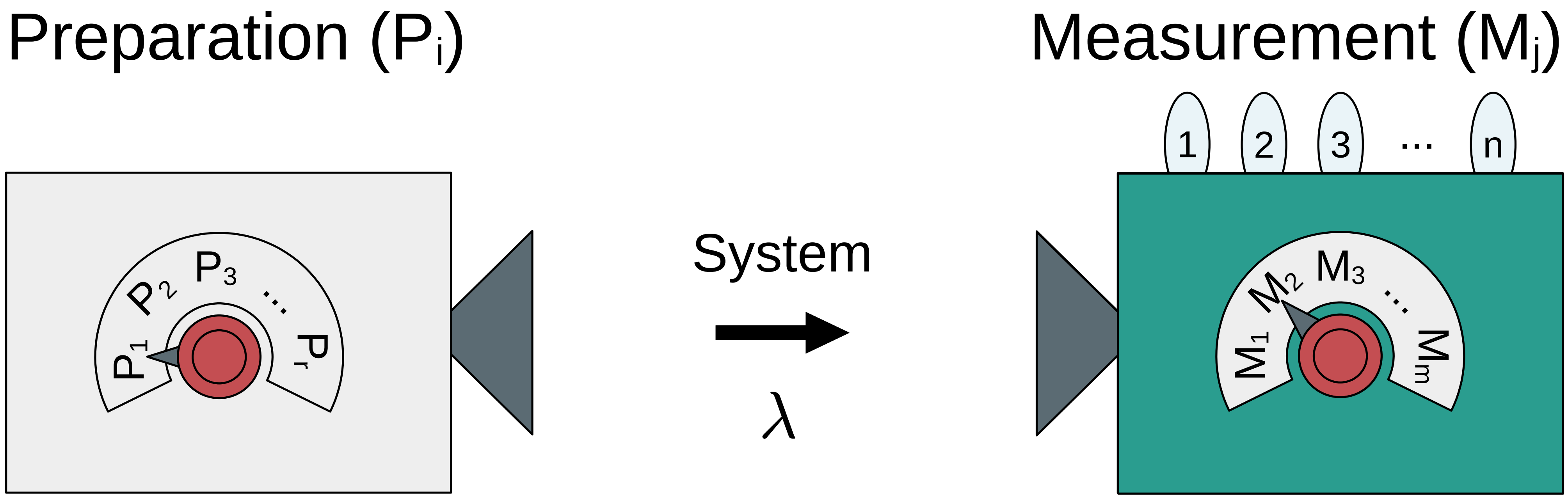
Outcome probabilities: $\mathrm{Prob}(k|P, M)$

Preparation and Measurement

Preparation (P_i)   Measurement (M_j)

Outcome probabilities: $\mathrm{Prob}(k|P_i, M_j)$

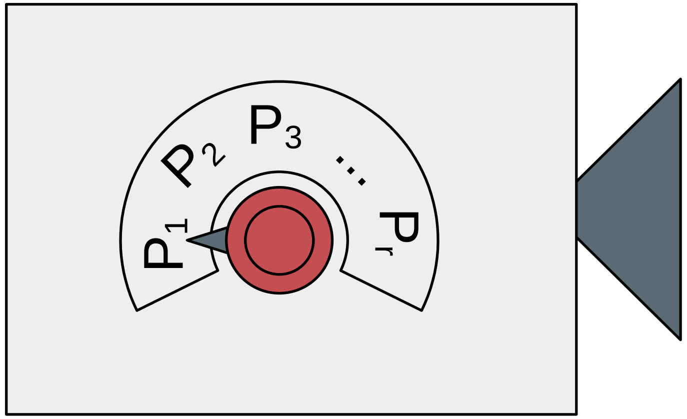
A classical experiment



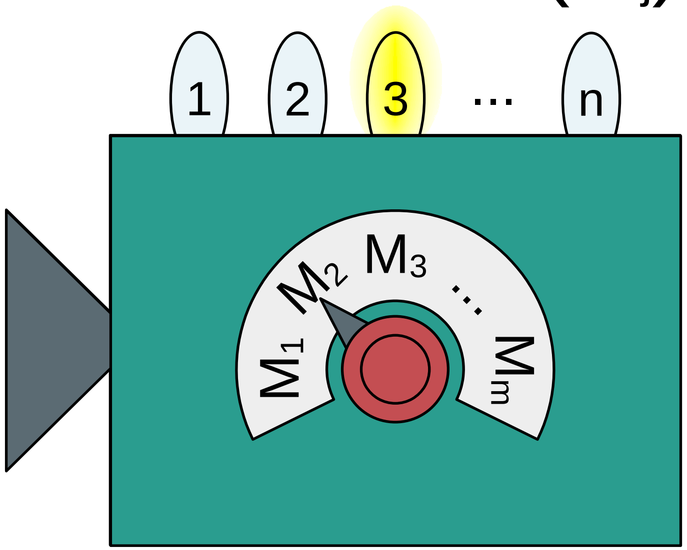
Defined set of properties: $\lambda = (\lambda_1, \lambda_2, \lambda_3, \ldots, \lambda_m)$

A classical experiment

Preparation (P_i)



Measurement (M_j)



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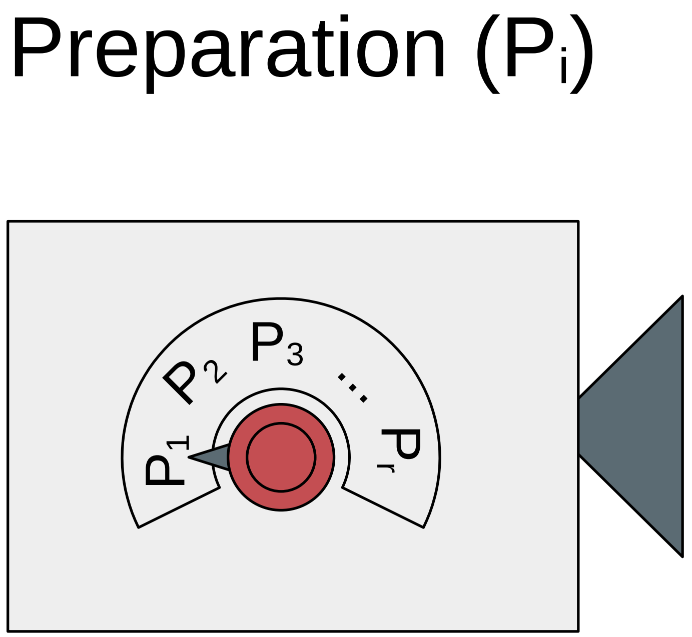
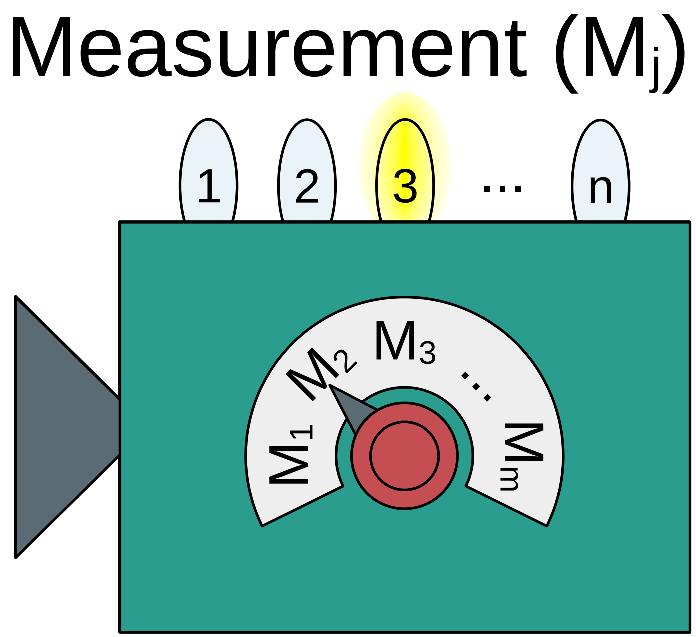
A classical experiment





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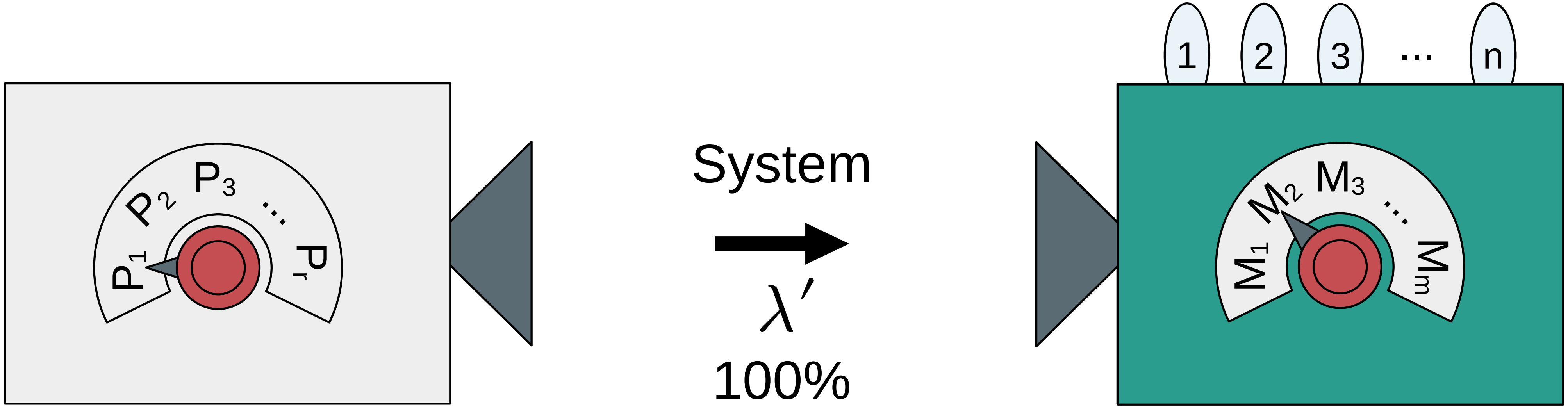
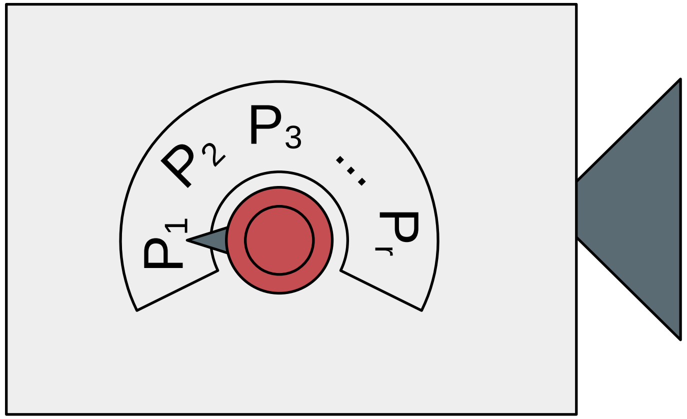
A classical experiment

Preparation (P_i) Measurement (M_j)

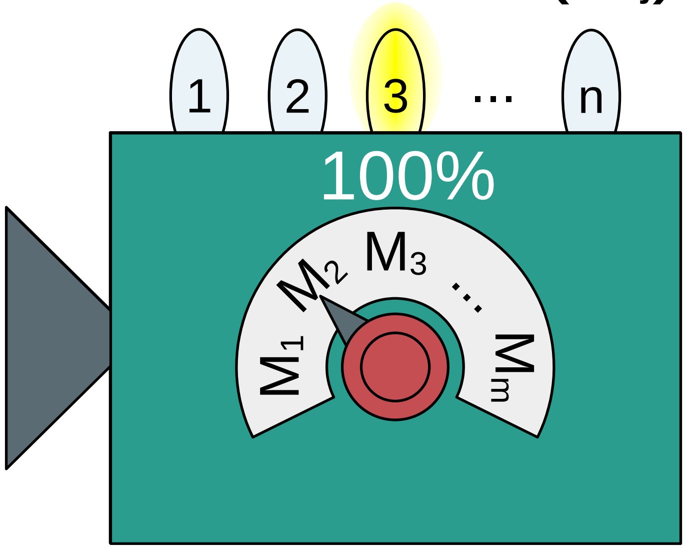


A classical experiment

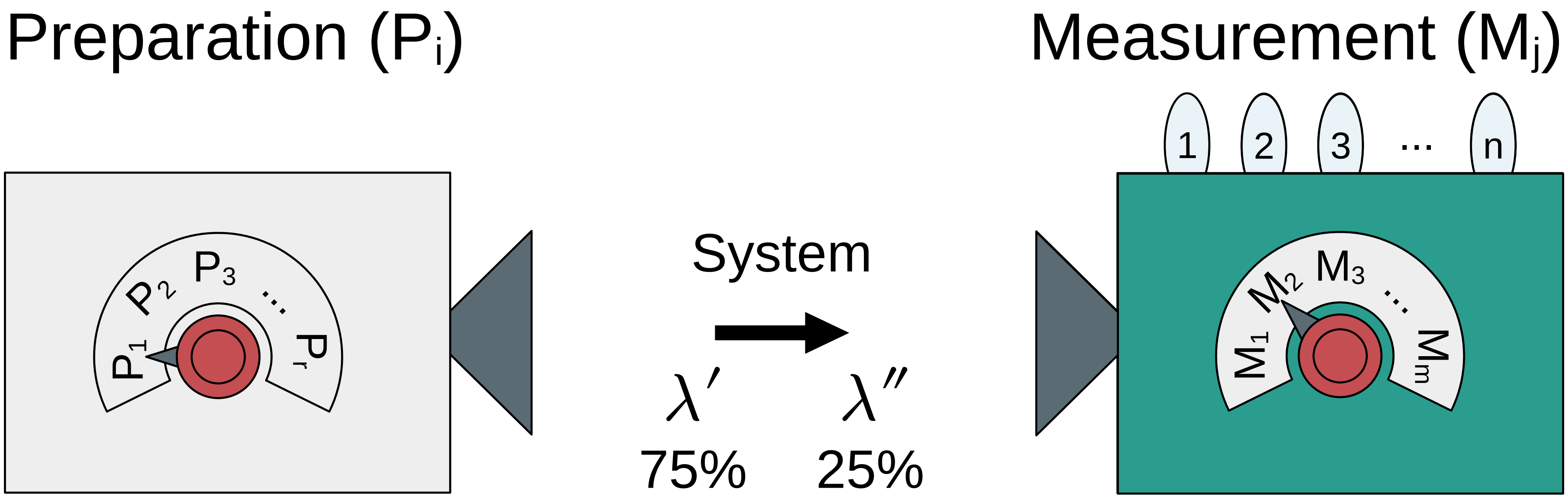
Preparation (P_i)



Measurement (M_j)

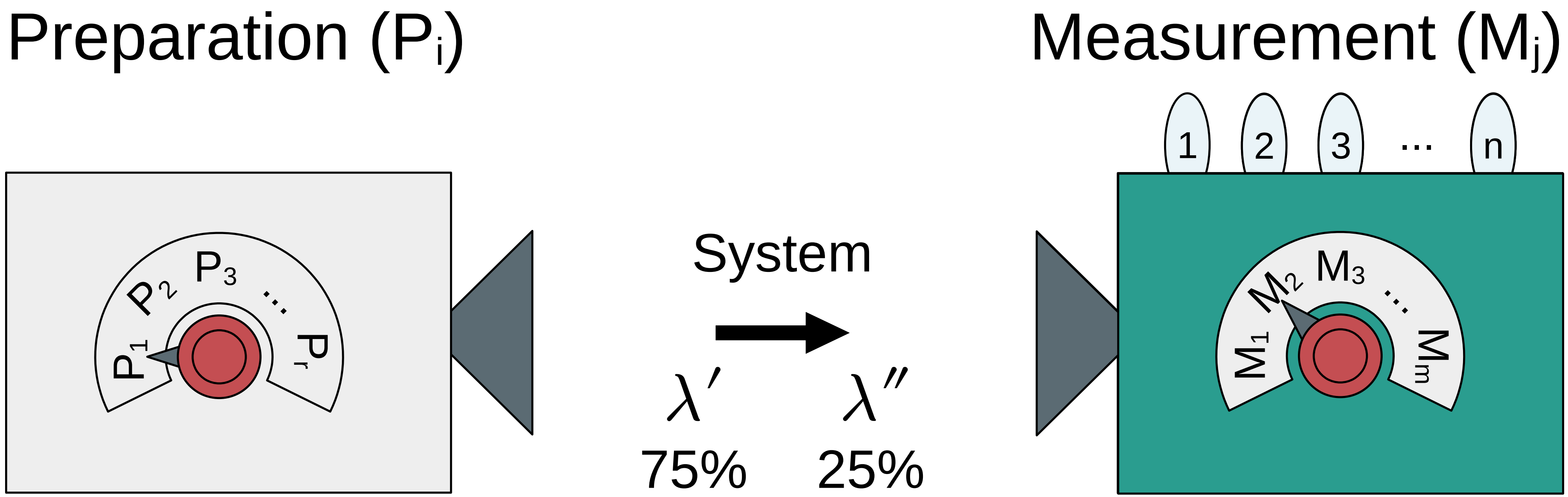


A classical experiment



A classical experiment

Probability distribution: $\mu(\lambda|P_i)$



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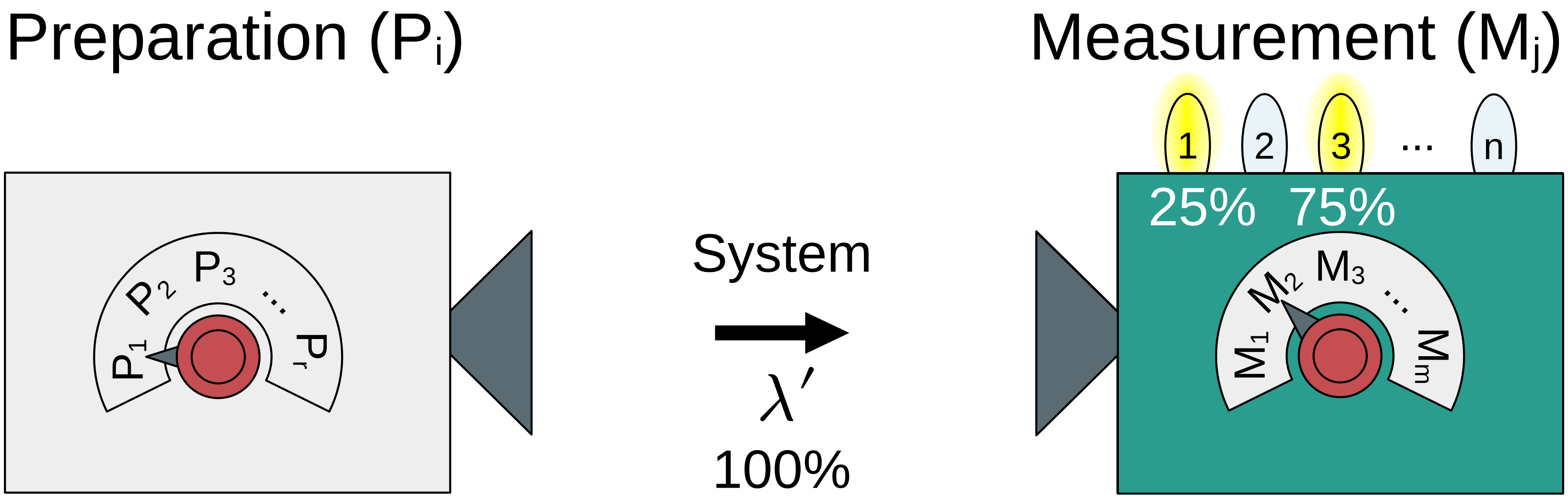
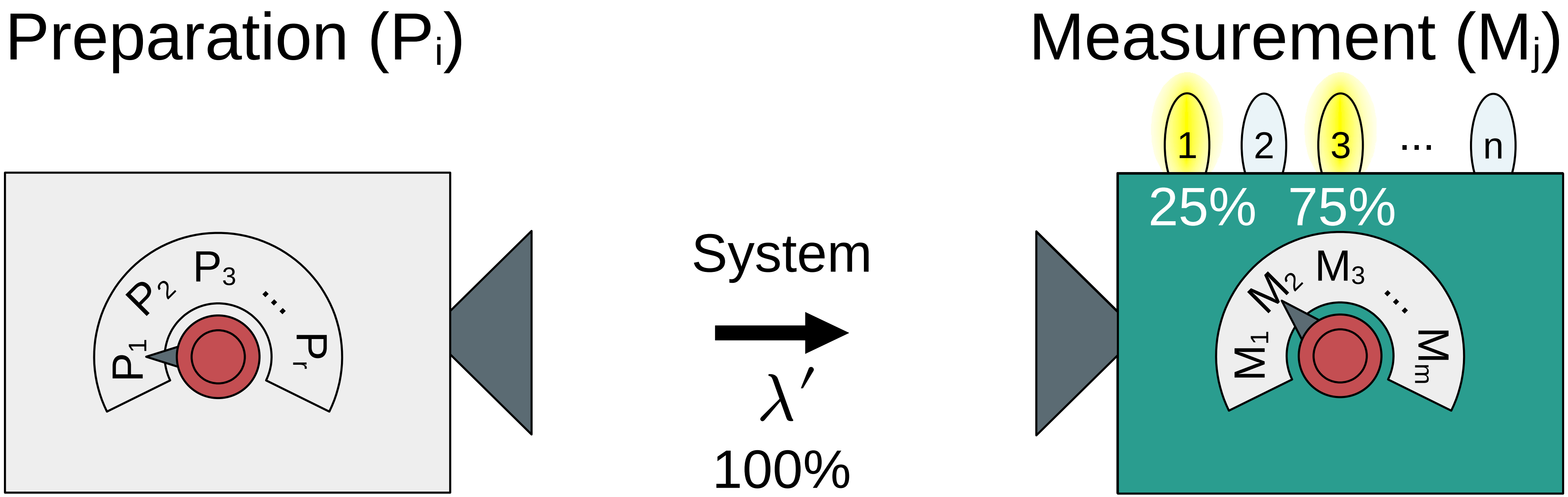


A classical experiment

Probability distribution: $\mu(\lambda|P_i)$

Response function: $\xi(k|M_j, \lambda)$



A classical experiment

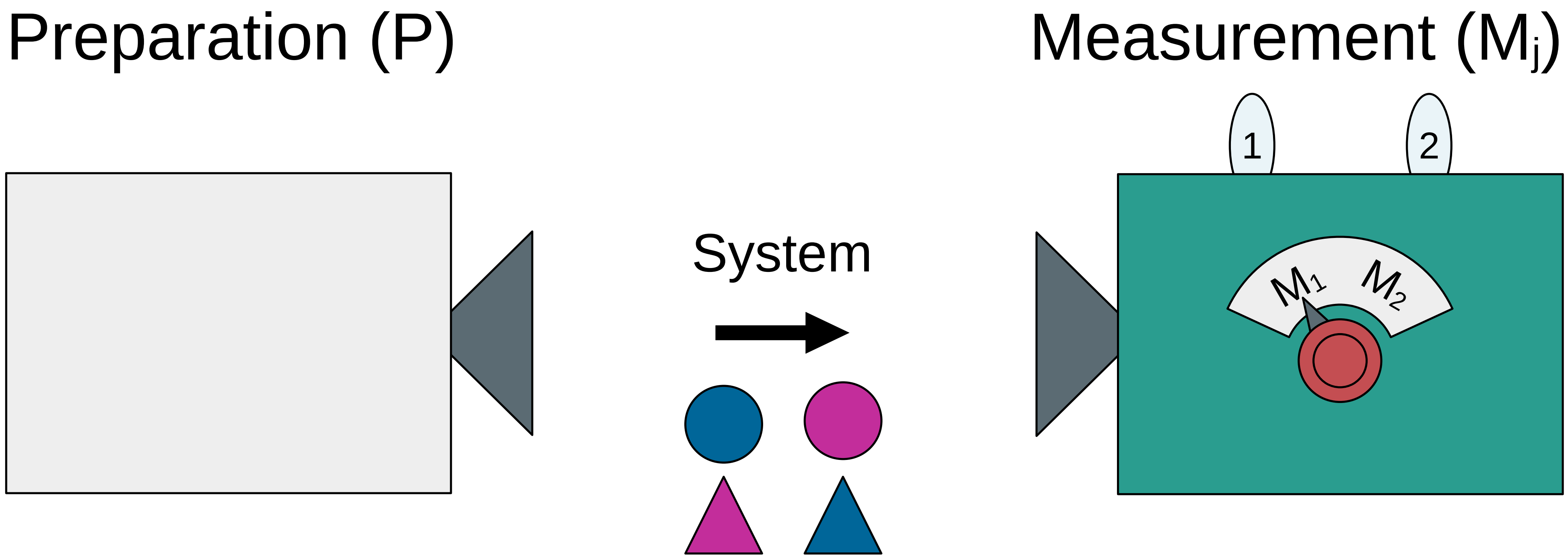
Probability distribution: $\mu(\lambda|P_i)$

Response function: $\xi(k|M_j, \lambda)$

Outcome probabilities:

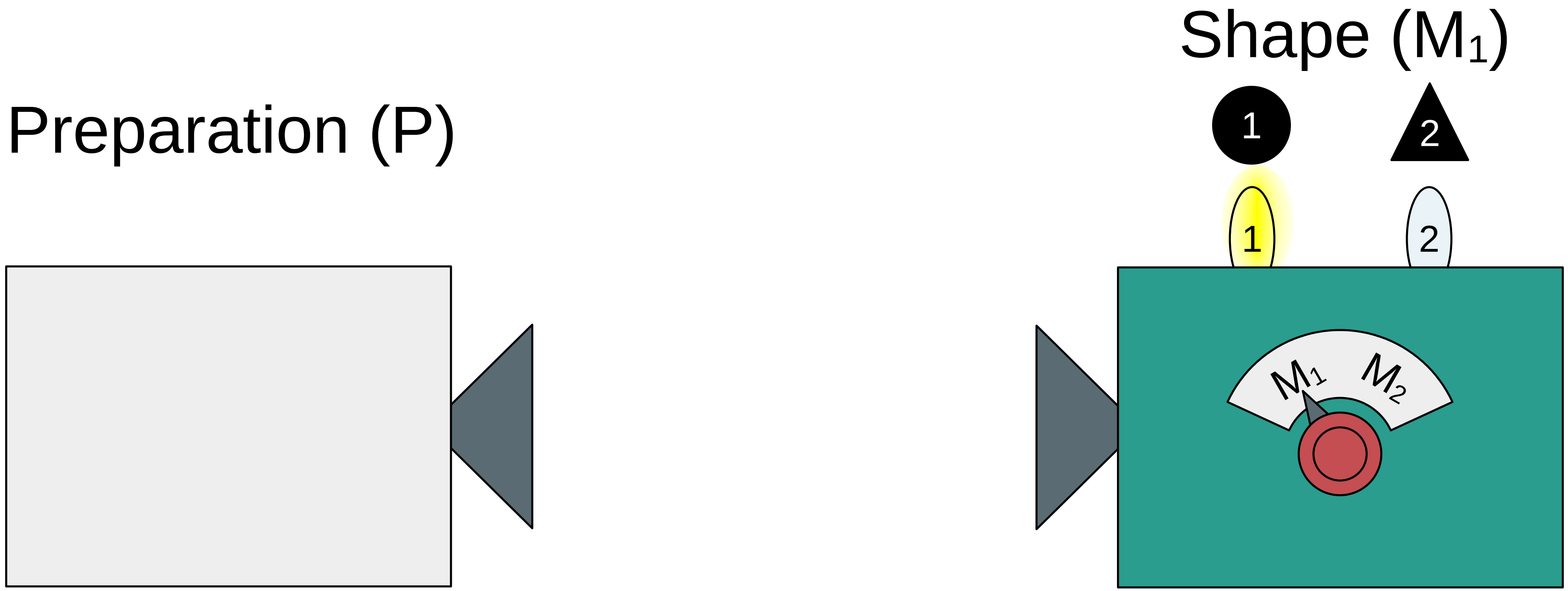
$$\mathrm{Prob}(k|P_i, M_j) = \sum_\lambda \mu(\lambda|P_i)\xi(k|M_j, \lambda)$$

A classical experiment



Two properties (shape, color): $\lambda = (\lambda_1, \lambda_2)$

A classical experiment



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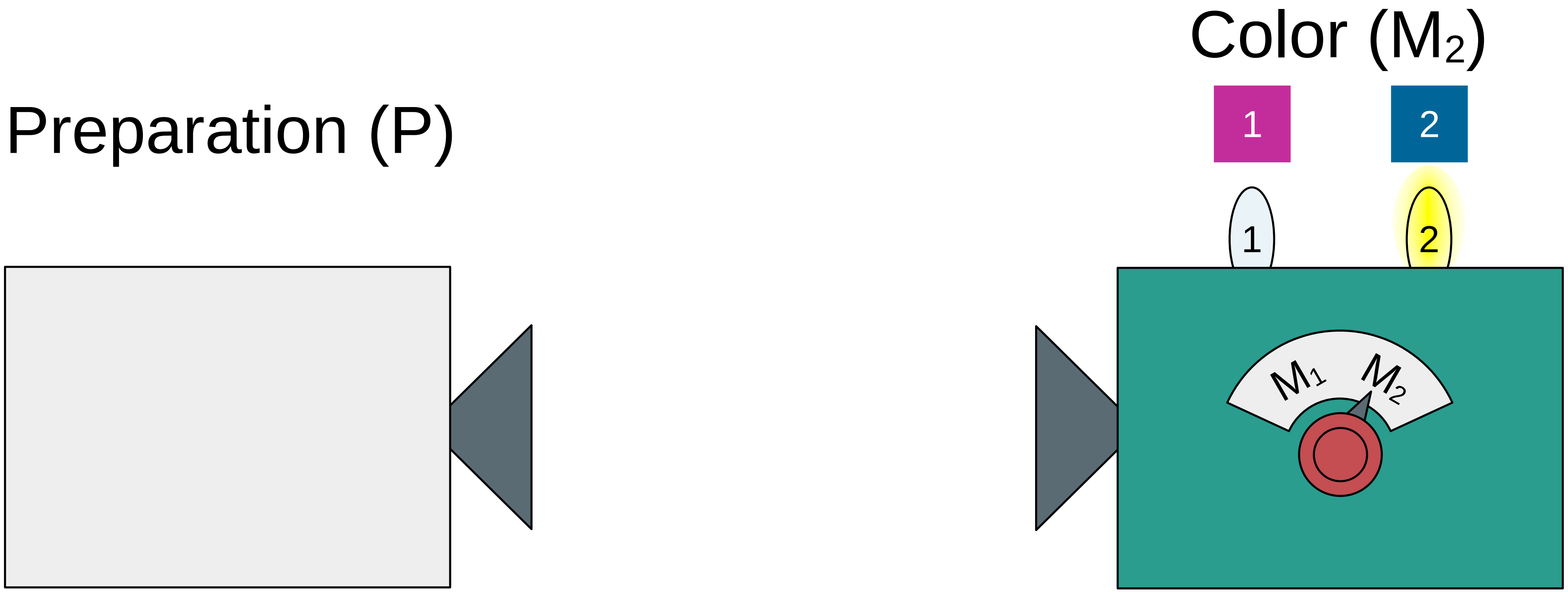


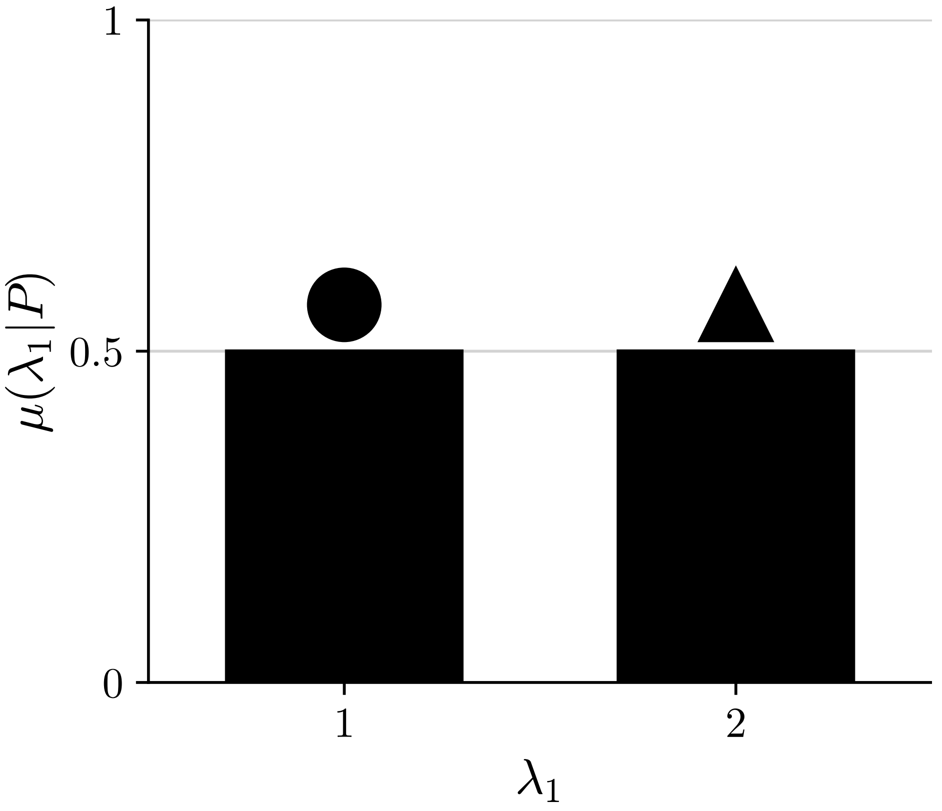
Two properties (shape, color): $\lambda = (\lambda_1, \lambda_2)$

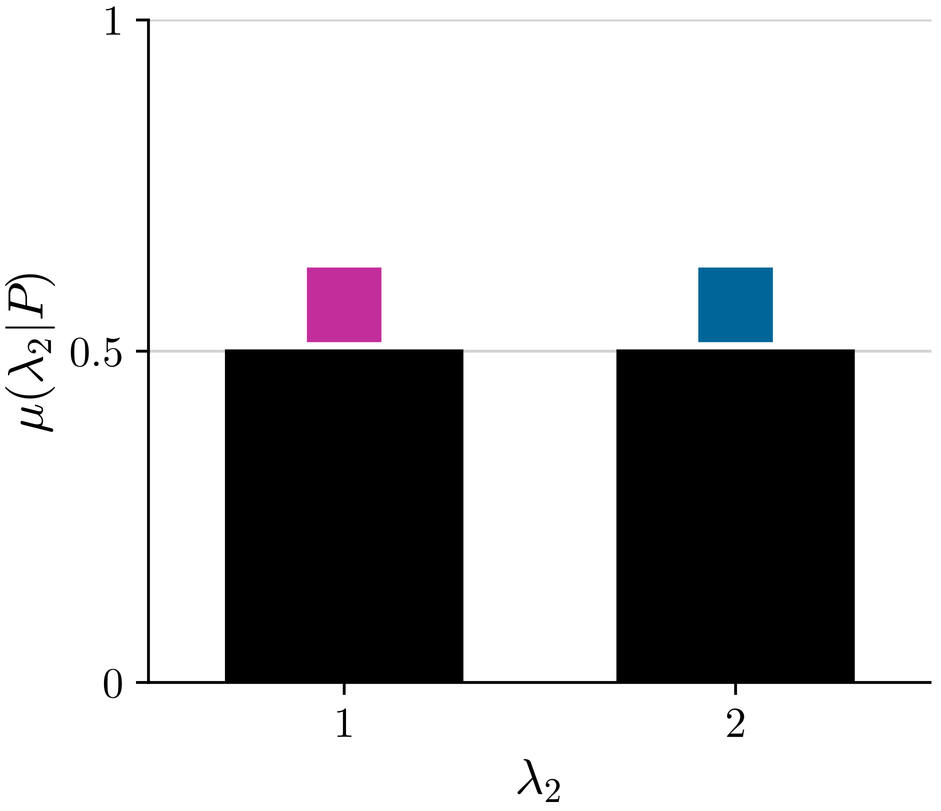
A classical experiment

50% 50%

Outcome probabilities:

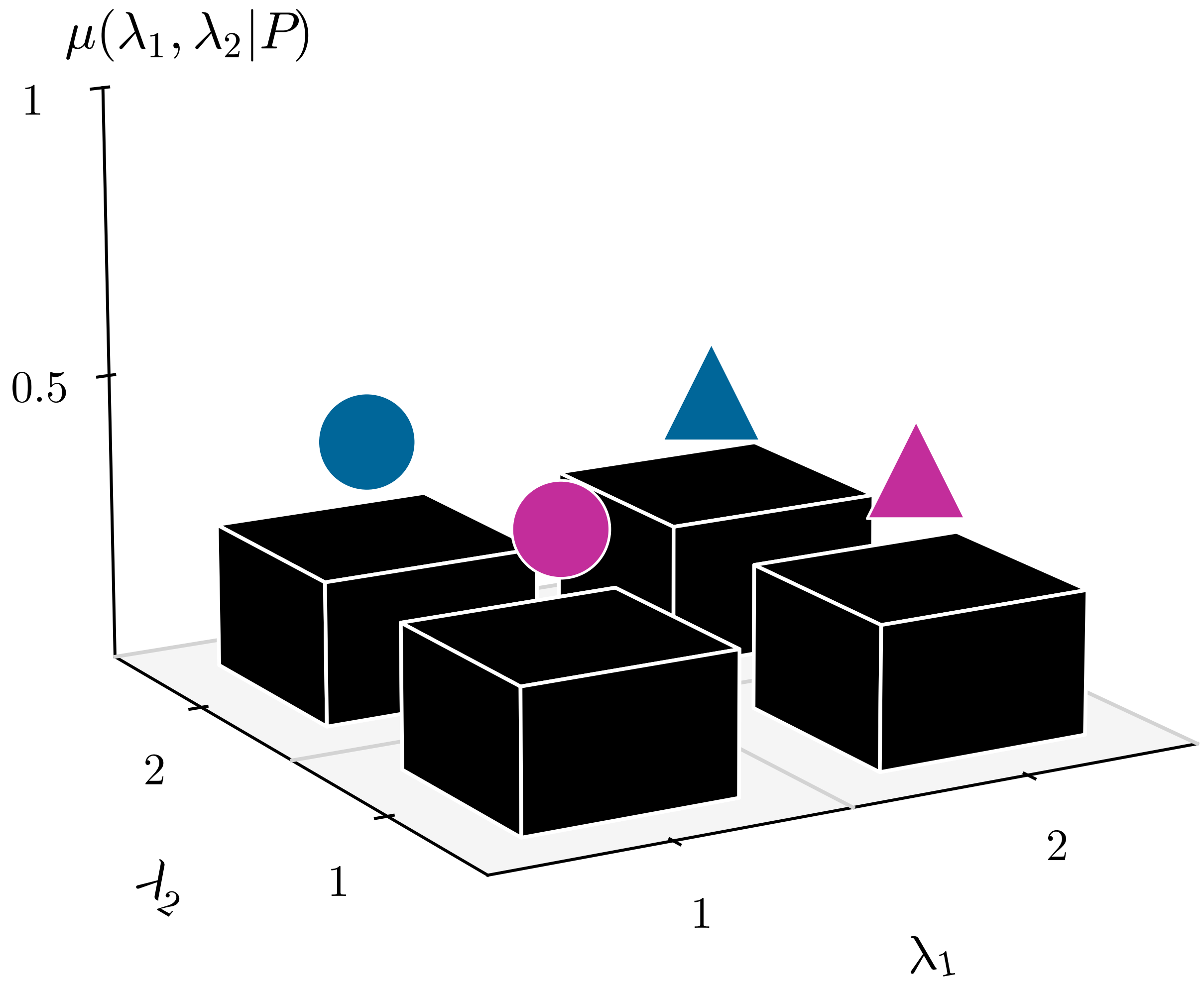
50% 50%

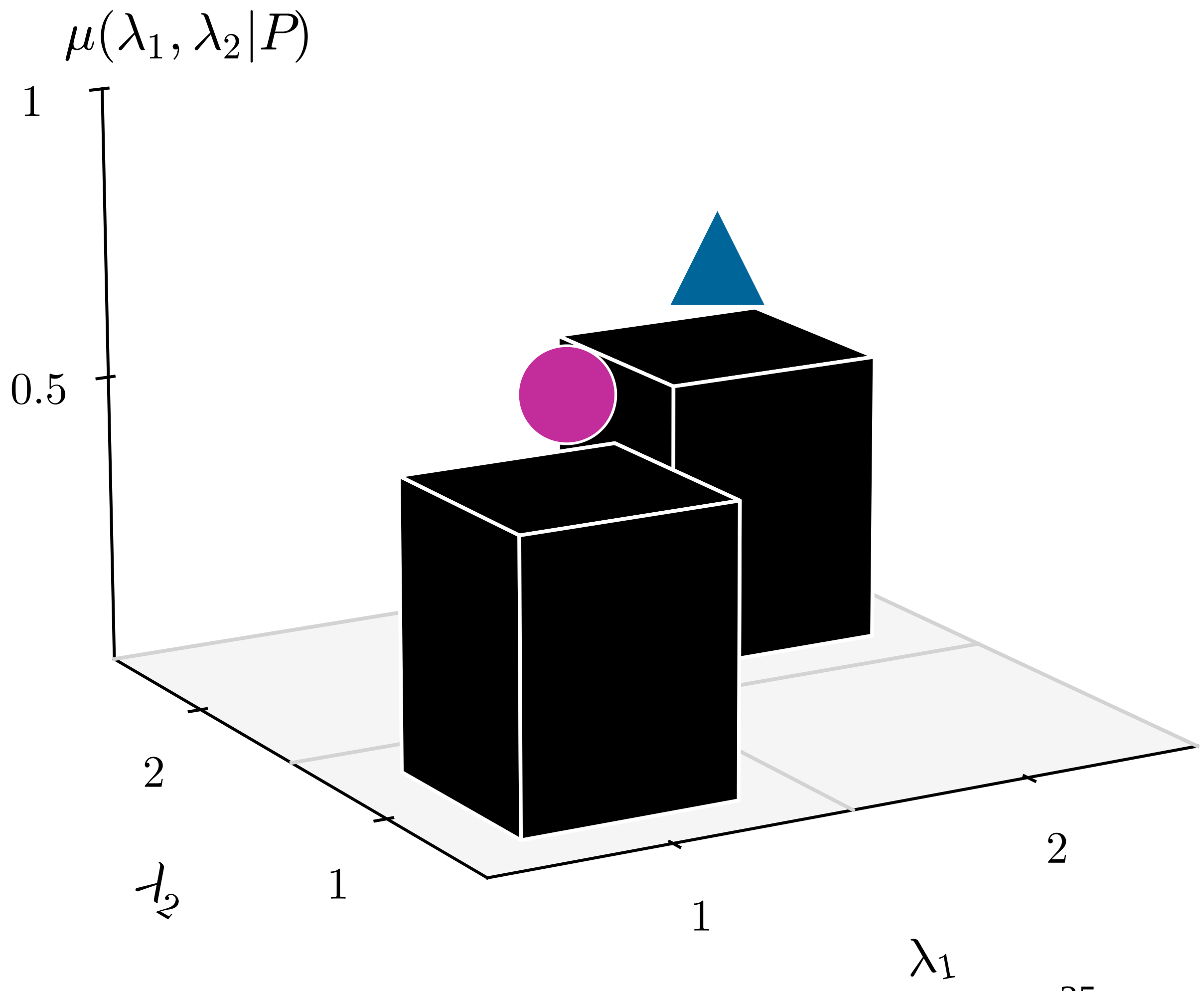




A classical experiment

Probability distribution: $\mu(\lambda|P)$





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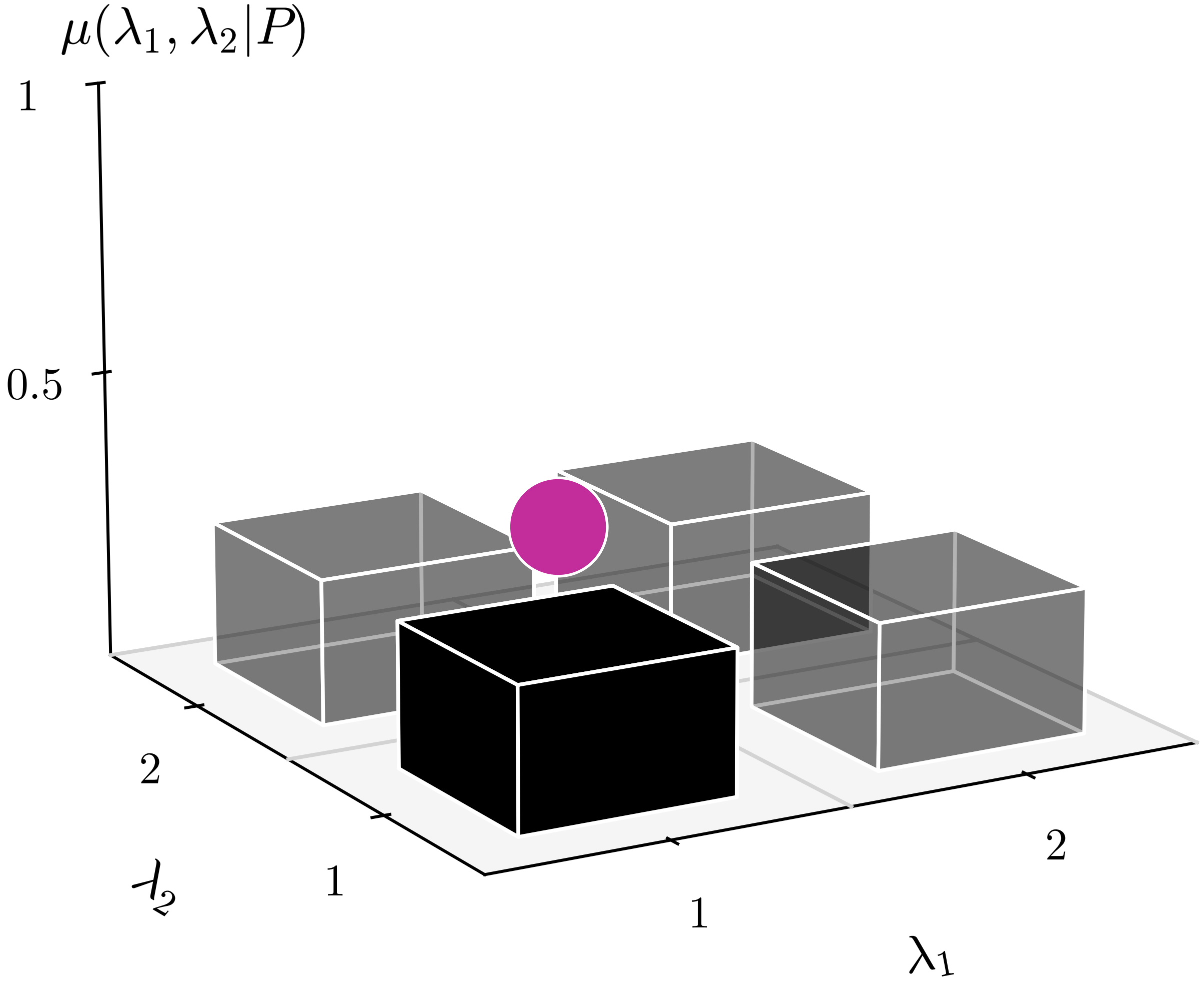
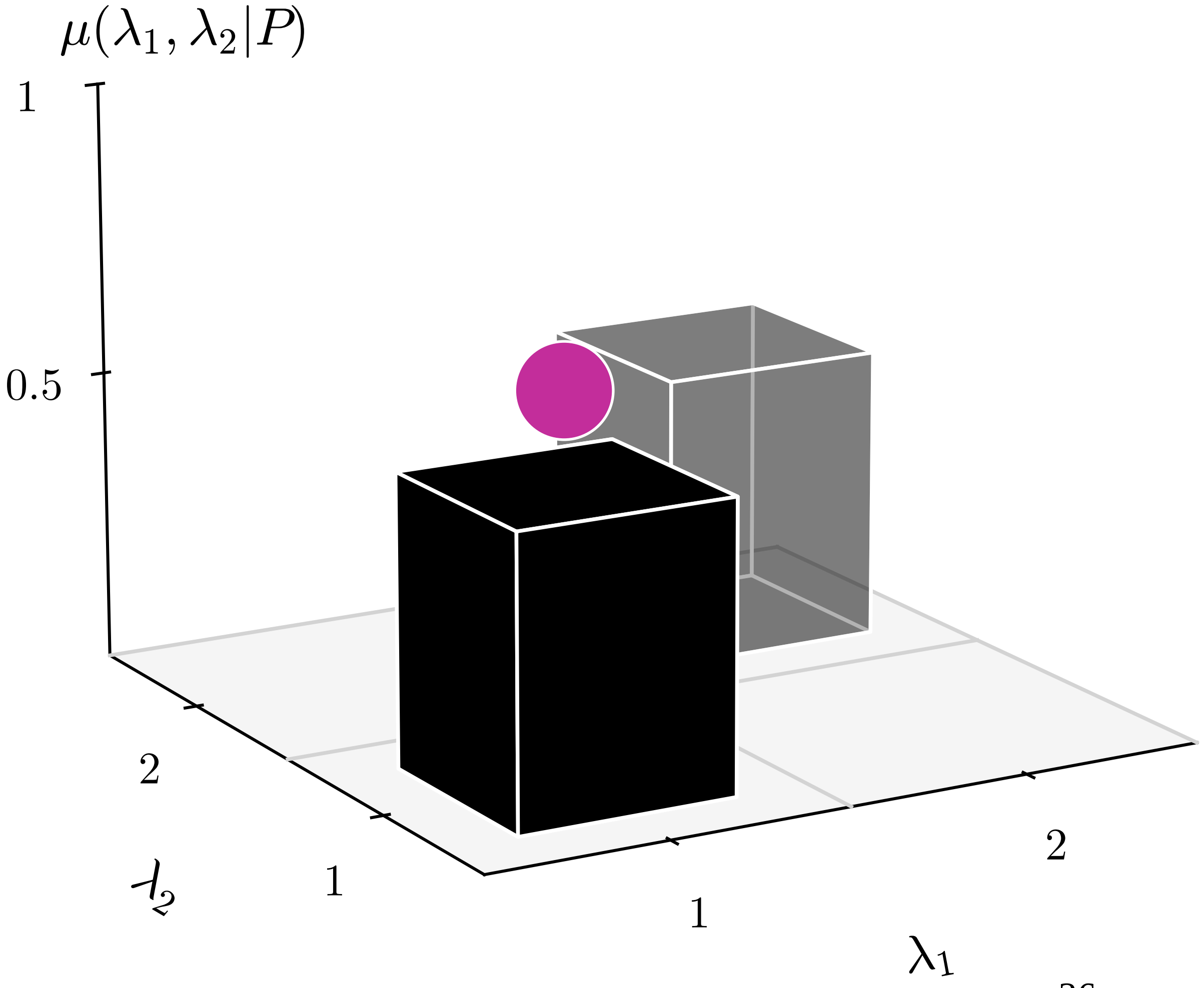
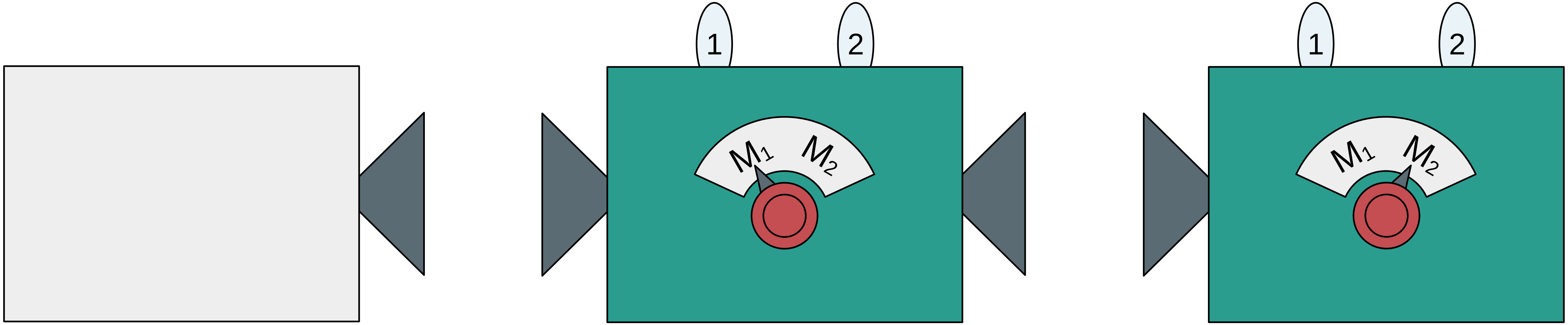


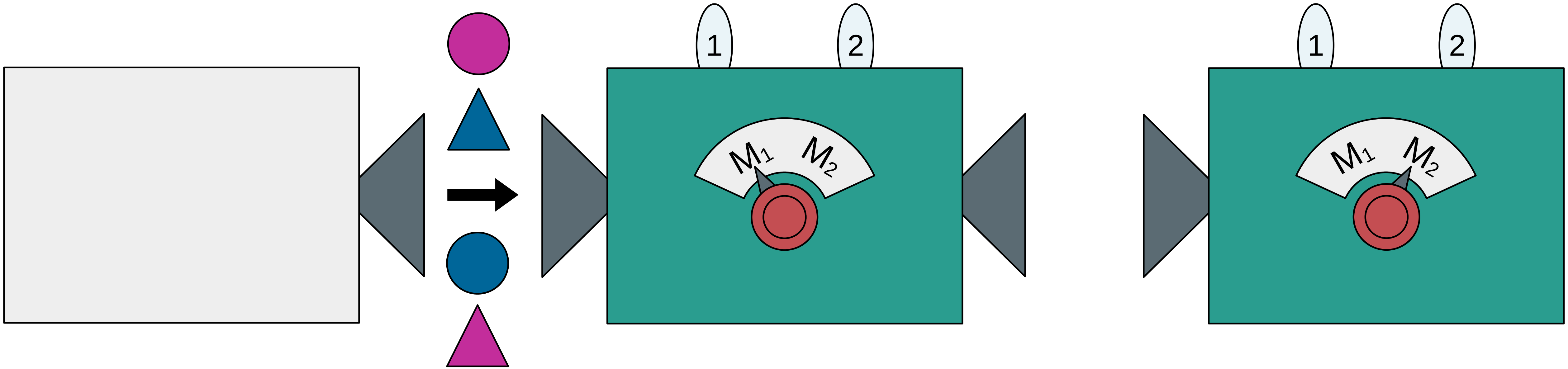


A classical experiment

Sequential (or simultaneous) measurement



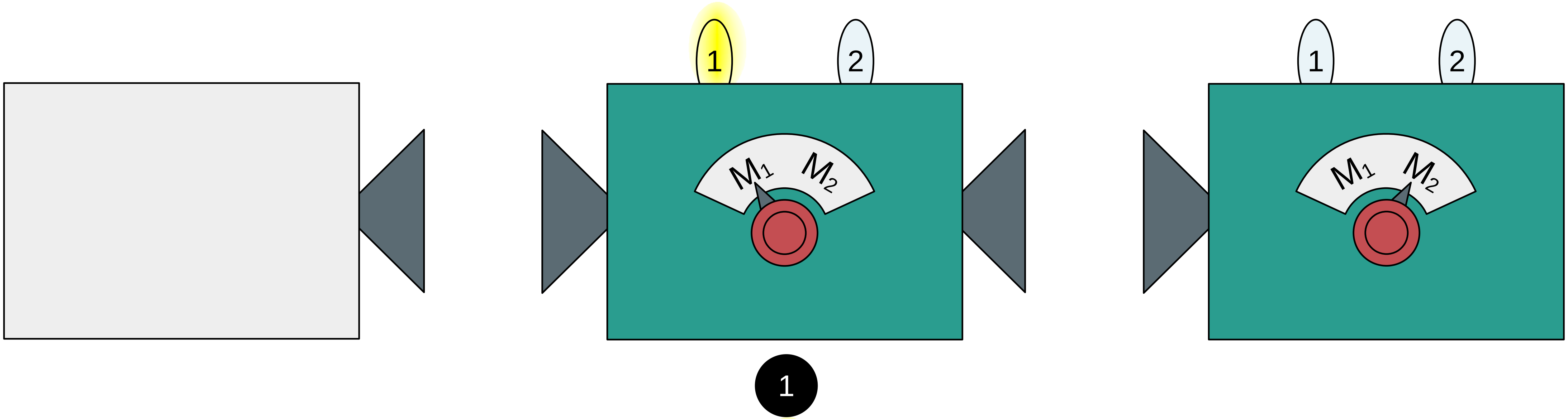
Sequential (or simultaneous) measurement



$$\lambda = (\lambda_1, \lambda_2)$$

A classical experiment

Sequential (or simultaneous) measurement



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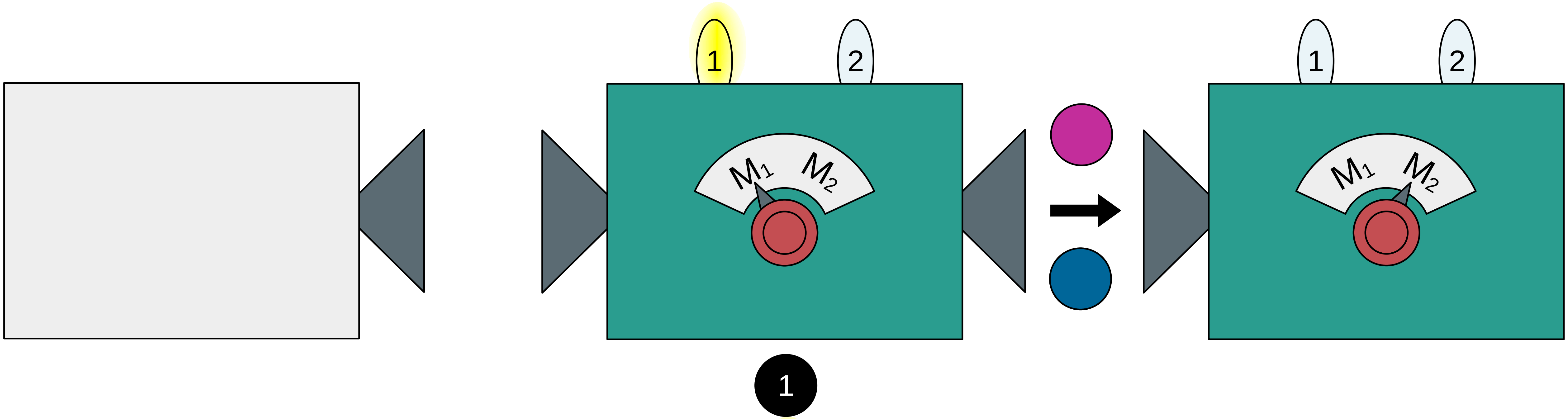
A classical experiment

Sequential (or simultaneous) measurement



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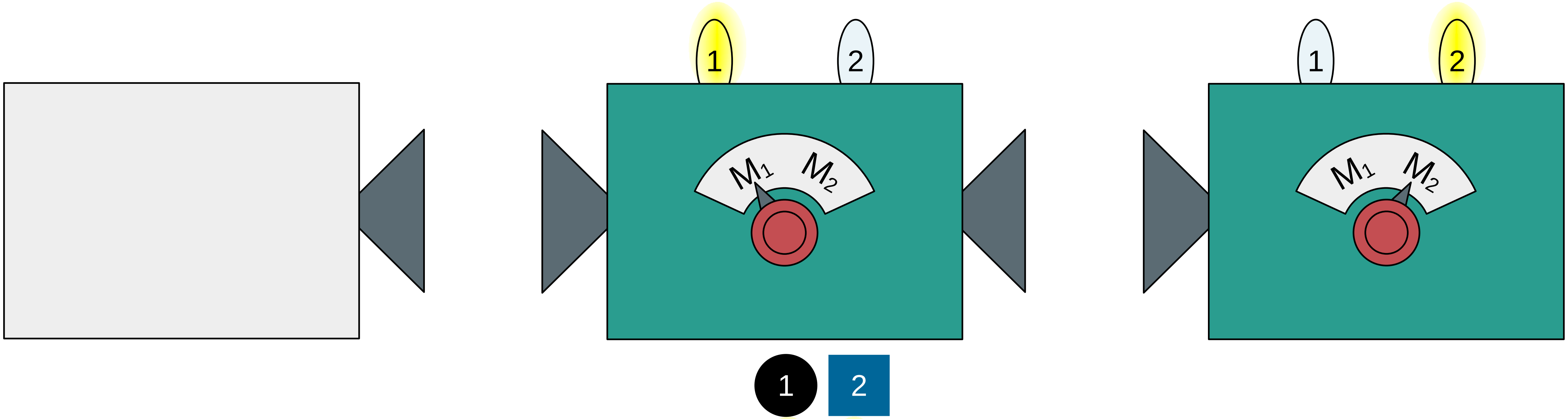
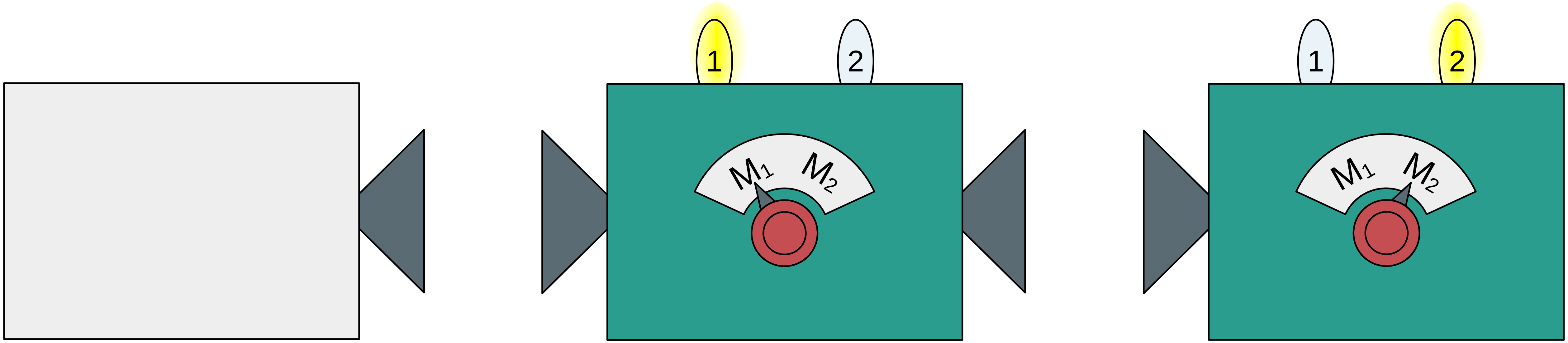
Sequential (or simultaneous) measurement



$$\lambda = (\lambda_1, \lambda_2)$$

Sequential (or simultaneous) measurement

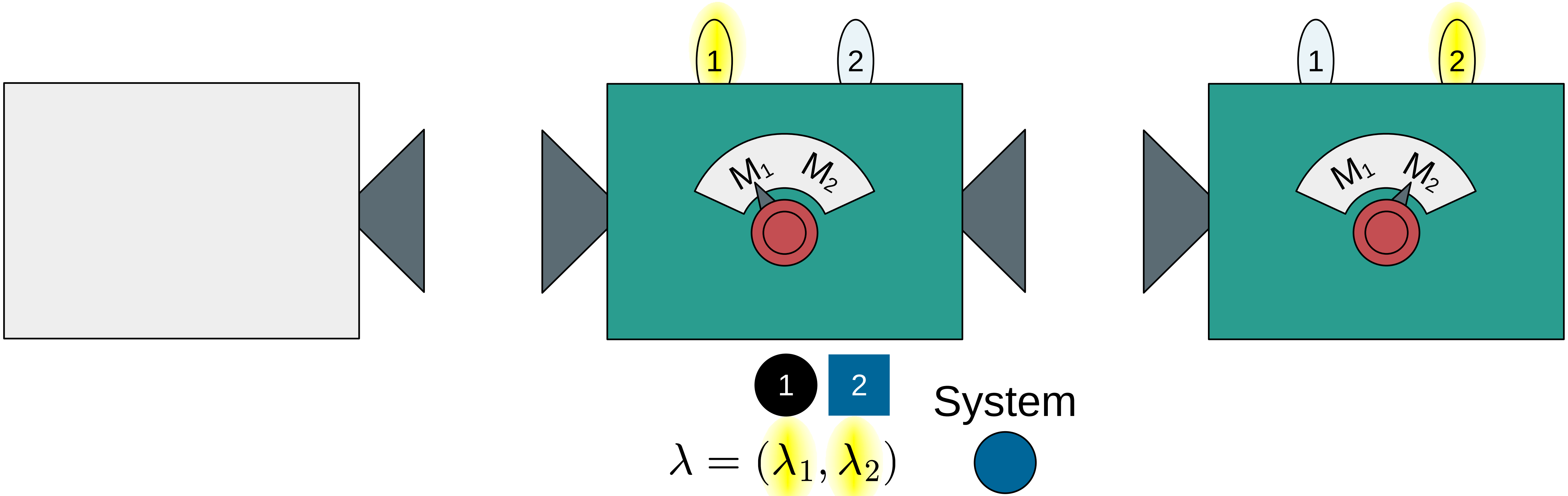


1 2 System

$$\lambda = (\lambda_1, \lambda_2)$$

A classical experiment

Sequential (or simultaneous) non-disturbing measurement



$\lambda = (\lambda_1, \lambda_2)$

A classical experiment

Outcome probabilities for non-disturbing measurements:

$$\mathrm{Prob}(l, m, \ldots | P, M_i, M_j, \ldots) = \sum_{\lambda} \mu(\lambda|P)\xi(l|M_i, \lambda)\xi(m|M_j, \lambda)\ldots$$

A classical experiment

Outcome probabilities for non-disturbing measurements:

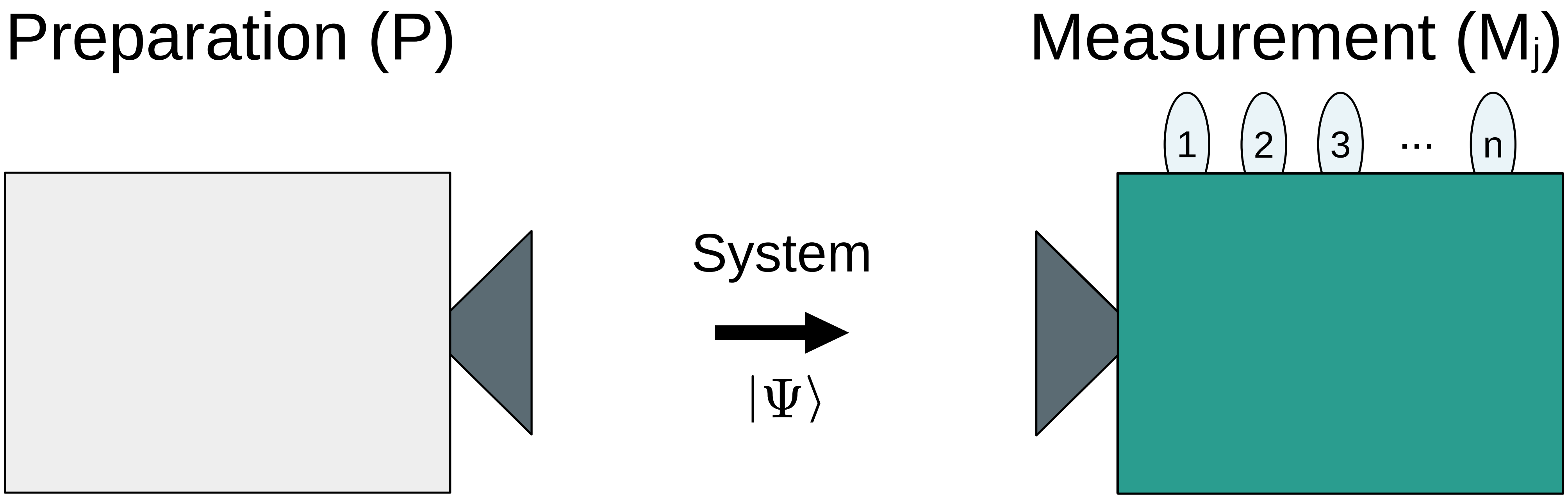
$$\mathrm{Prob}(\underbrace{l, m, \ldots}_{k} | P, \underbrace{M_i, M_j, \ldots}_{M}) = \sum_{\lambda} \mu(\lambda|P) \underbrace{\xi(l|M_i, \lambda)\xi(m|M_j, \lambda) \ldots}_{\xi(k|M,\lambda)}$$

A classical experiment

Outcome probabilities for non-disturbing measurements:

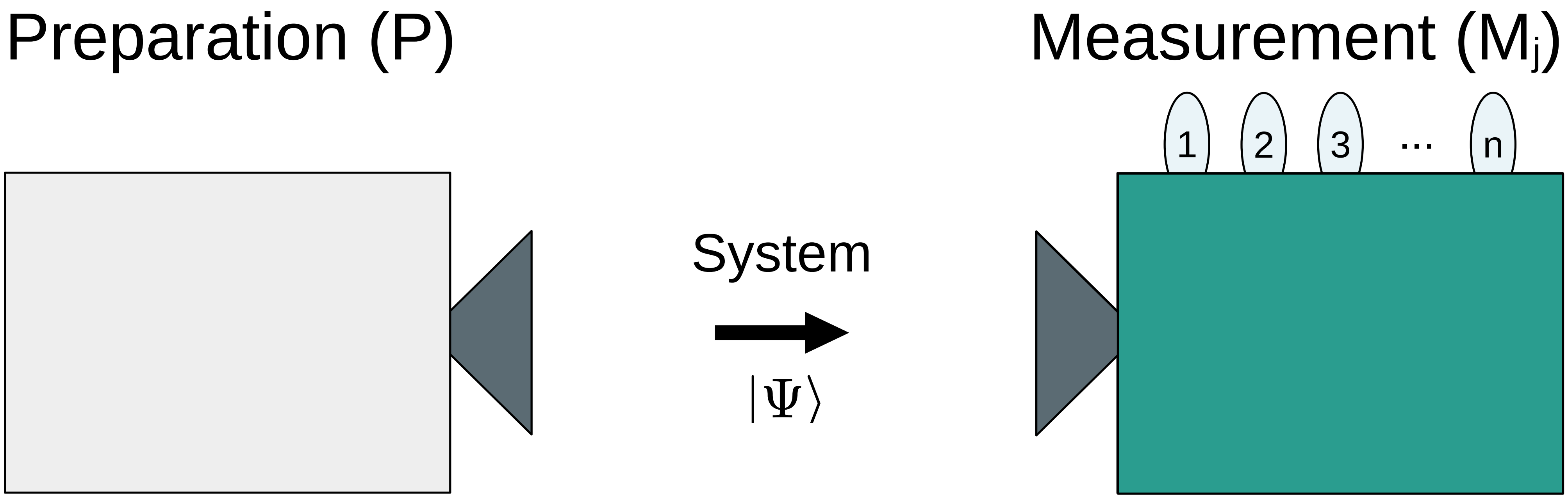
$$\mathrm{Prob}(k|P,M) = \sum_{\lambda} \mu(\lambda|P)\xi(k|M,\lambda)$$

A quantum experiment



Set of undefined properties: $|\Psi\rangle = (|\psi_1\rangle, |\psi_2\rangle, |\psi_3\rangle, \ldots)$

A quantum experiment

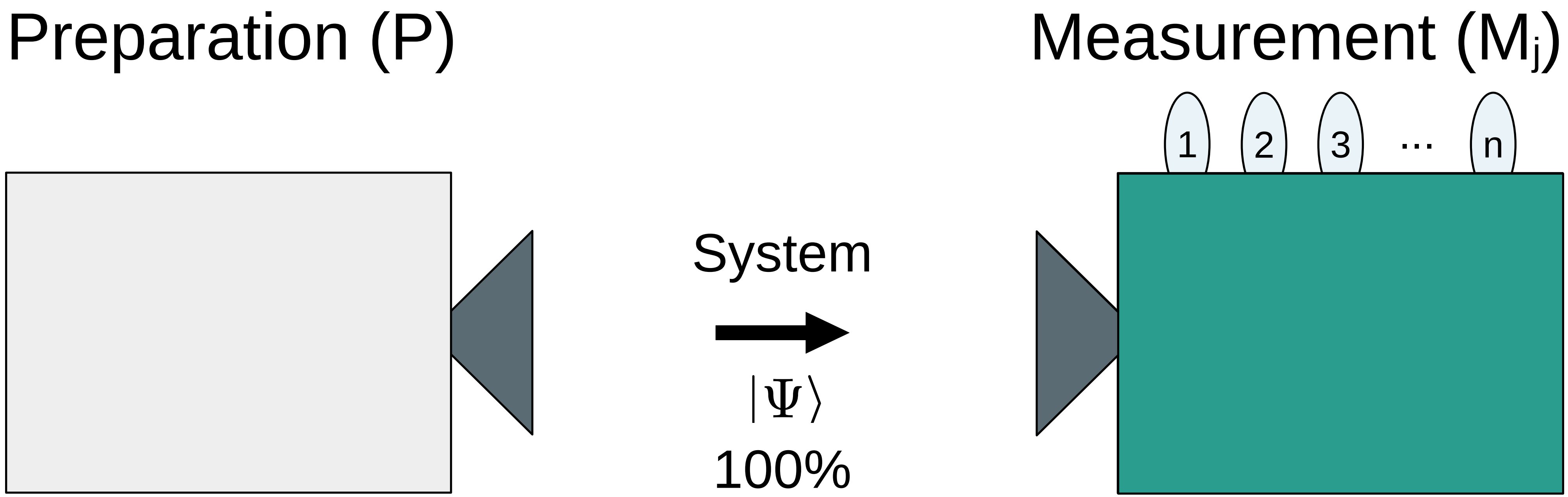


Set of undefined properties: $|\Psi\rangle = (|\psi_1\rangle, |\psi_2\rangle, |\psi_3\rangle, \ldots)$

Superposition: $|\psi_j\rangle = a_1|1\rangle + a_2|2\rangle + a_3|3\rangle + \ldots$

A quantum experiment

Pure state: $|\Psi\rangle$



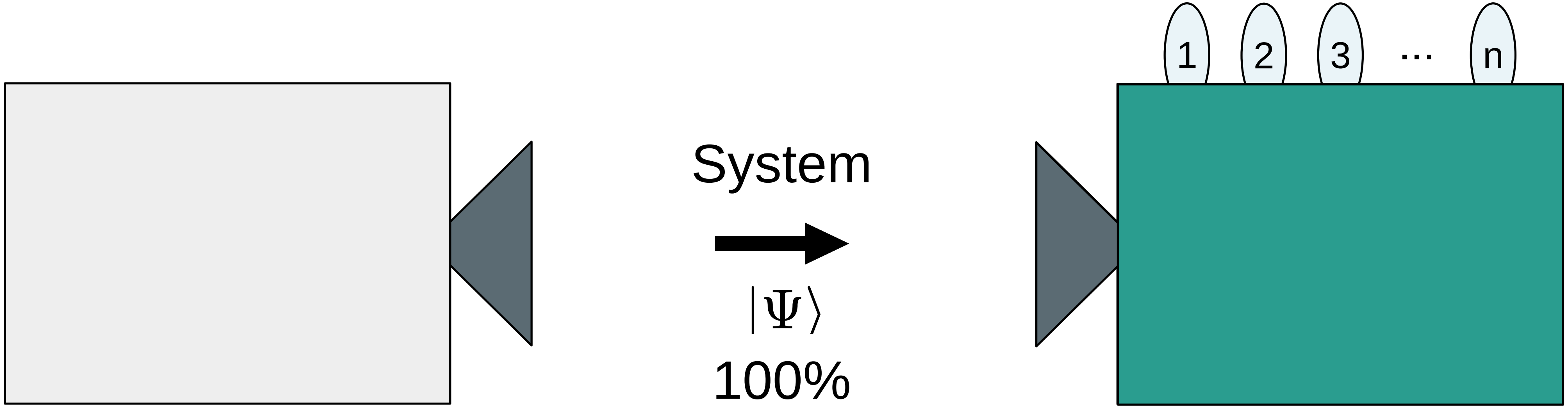
A quantum experiment

Pure state: $|\Psi\rangle$

Projective measurement: $\hat{\Pi}_{k|M}$

Preparation (P)

Measurement (M_j)



A quantum experiment

Pure state: $|\Psi\rangle$

Projective measurement: $\hat{\Pi}_{k|M}$

Outcome probabilities: $\mathrm{Prob}(k|P,M) = \langle\Psi|\hat{\Pi}_{k|M}|\Psi\rangle \leq 1$

A quantum experiment

Pure state: $|\Psi\rangle$

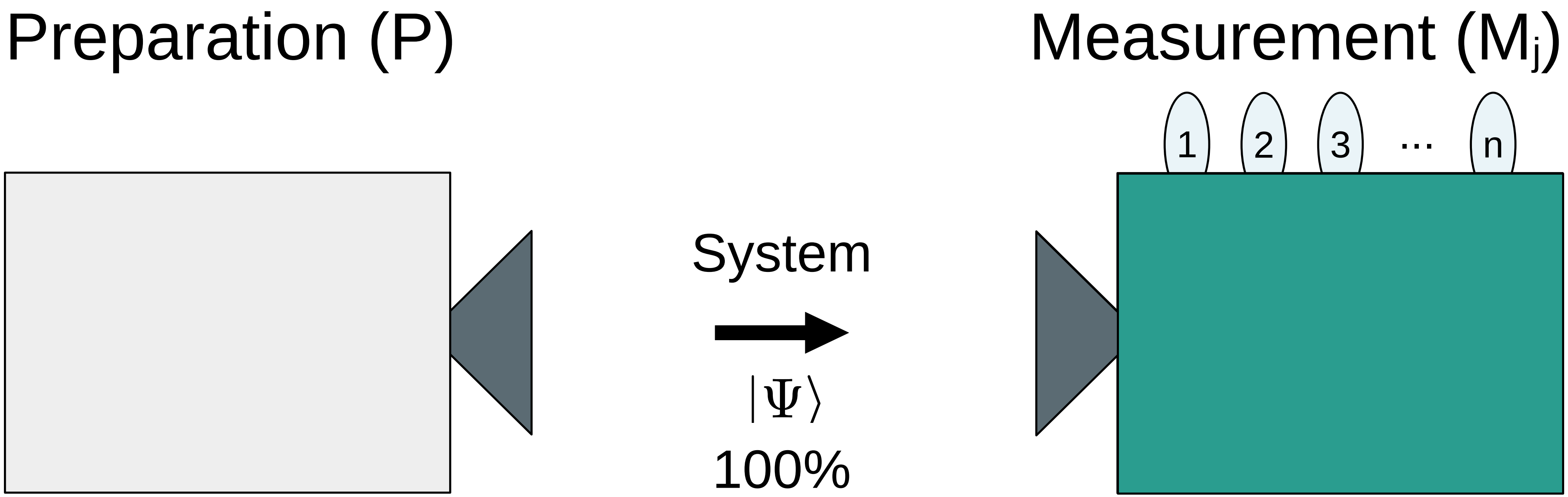
Projective measurement: $\hat{\Pi}_{k|M}$

Is quantum mechanics secretly classical?

Outcome probabilities: $\mathrm{Prob}(k|P,M) = \langle\Psi|\hat{\Pi}_{k|M}|\Psi\rangle \leq 1$

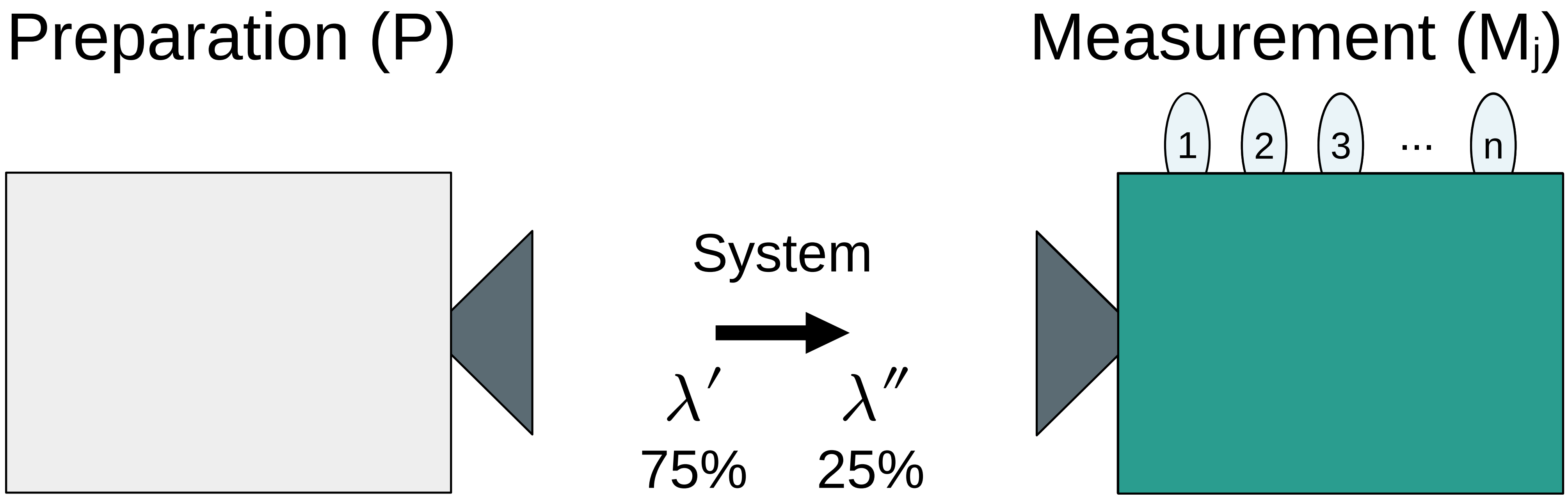
Is quantum mechanics classical?

Pure state: $|\Psi\rangle$



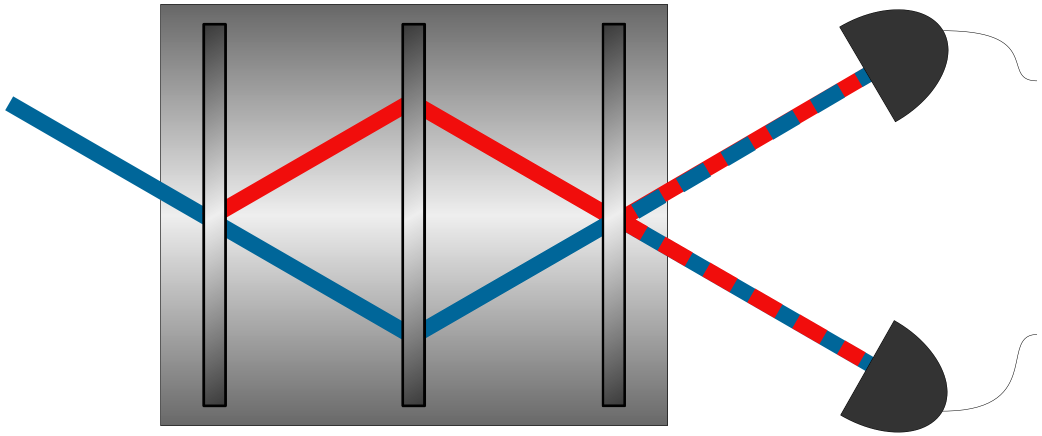
Is quantum mechanics classical?

Probability distribution: $\mu(\lambda|P_i)$



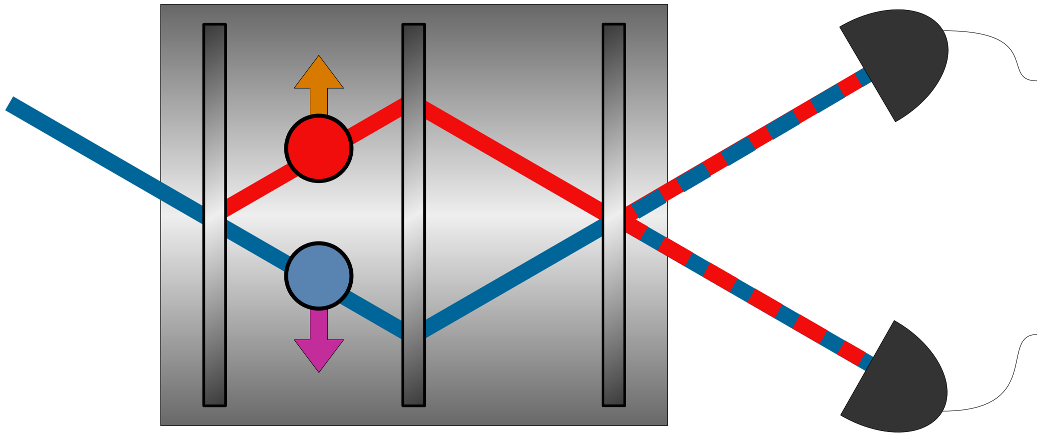
Neutron interferometry test of nonclassicality

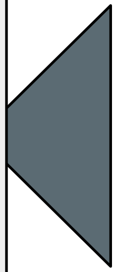
Neutron Interferometer:



Neutron interferometry test of nonclassicality

Neutron Interferometer:



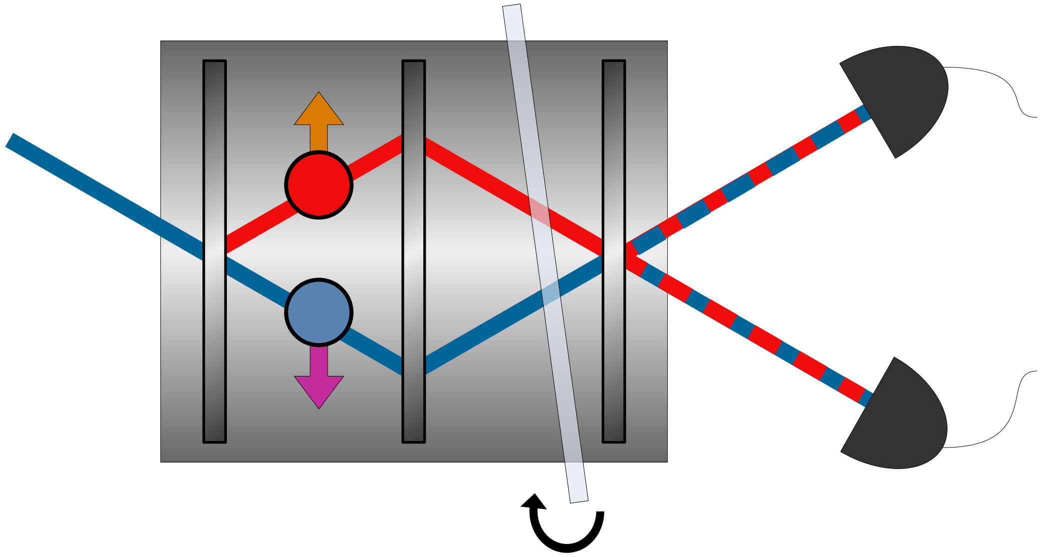


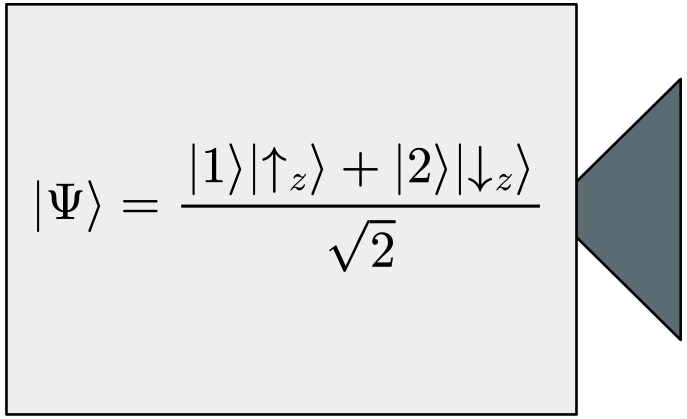
$$|\Psi\rangle = \frac{|1\rangle|\uparrow_z\rangle + |2\rangle|\downarrow_z\rangle}{\sqrt{2}}$$

Preparation (P)

Neutron interferometry test of nonclassicality

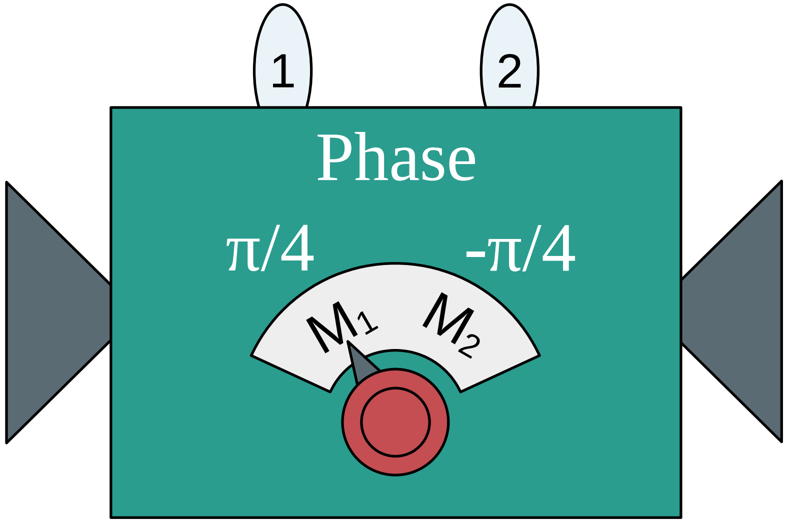
Neutron Interferometer:





$$|\Psi\rangle = \frac{|1\rangle|\uparrow_z\rangle + |2\rangle|\downarrow_z\rangle}{\sqrt{2}}$$

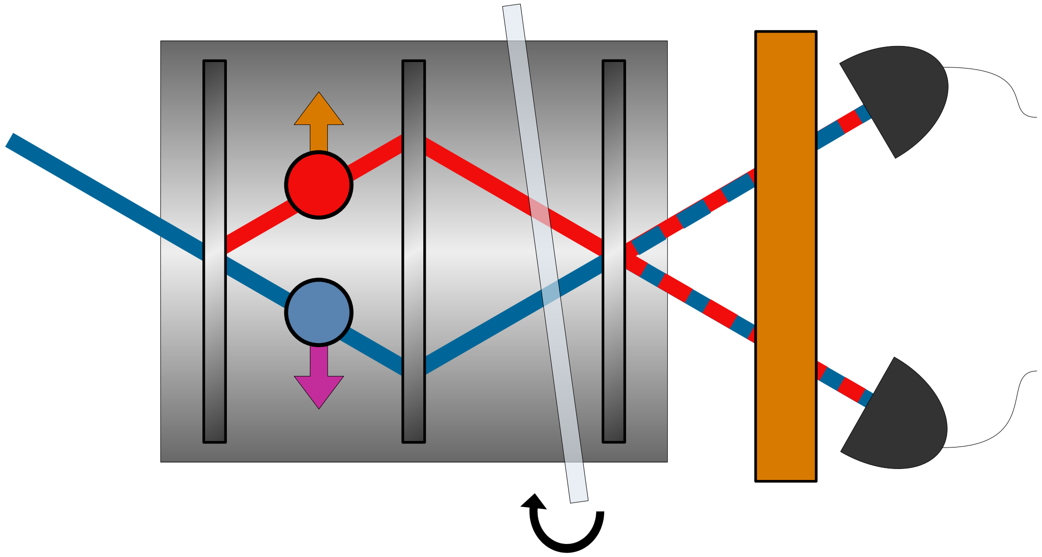
Preparation (P)

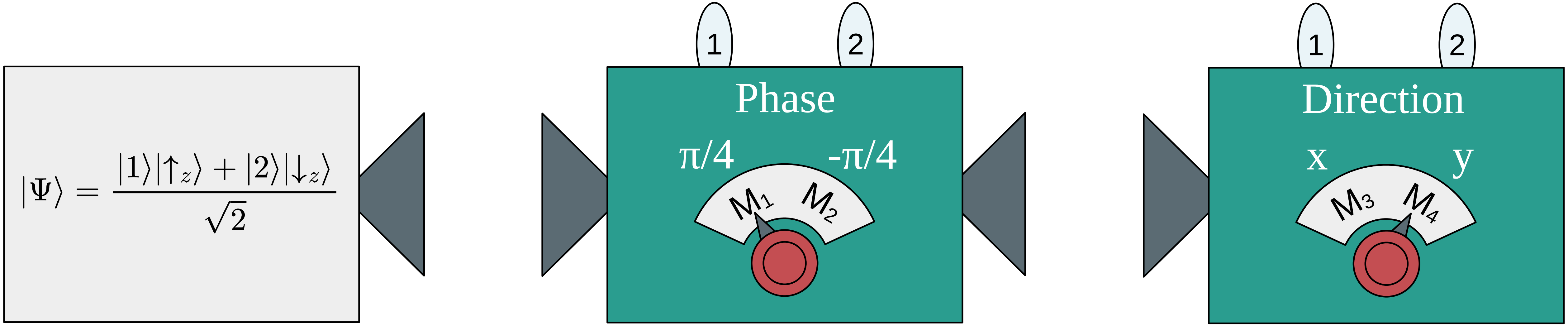


Path (M_1,M_2)

Neutron interferometry test of nonclassicality

Neutron Interferometer:



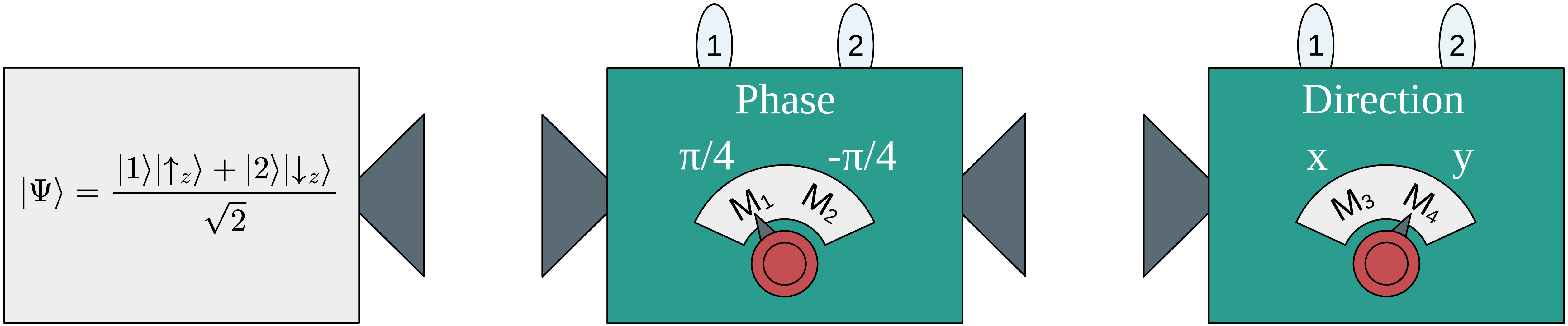


Preparation (P) Path (M_1,M_2) Spin (M_3,M_4)

Measurements:

M_1 M_2

M_3 M_4



Preparation (P) Path (M_1,M_2) Spin (M_3,M_4)

Measurements:

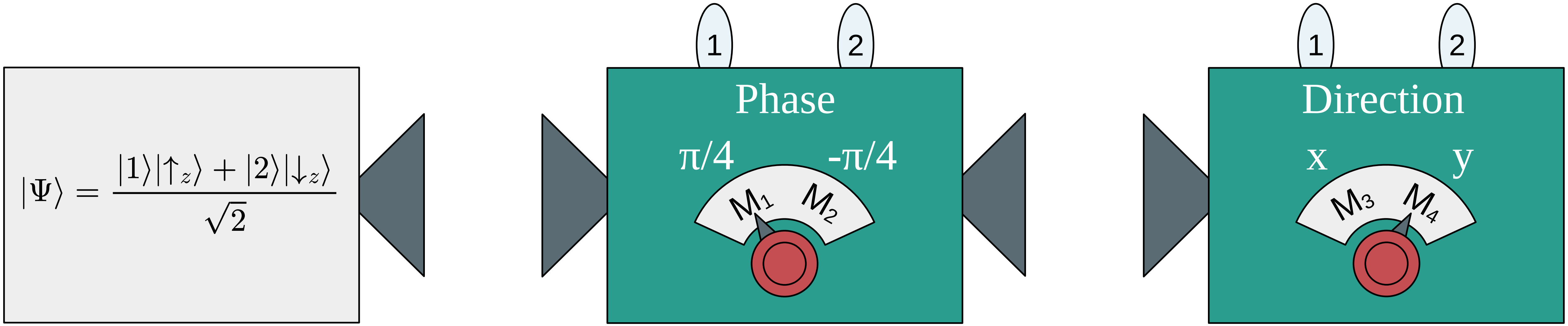
$M_1 \leftrightarrow M_2$

Disturbing

$\leftrightarrow$

$$\left[\hat{\Pi}_{k|M_i}, \hat{\Pi}_{l|M_j}\right] \neq 0$$

$M_3 \leftrightarrow M_4$



Preparation (P)

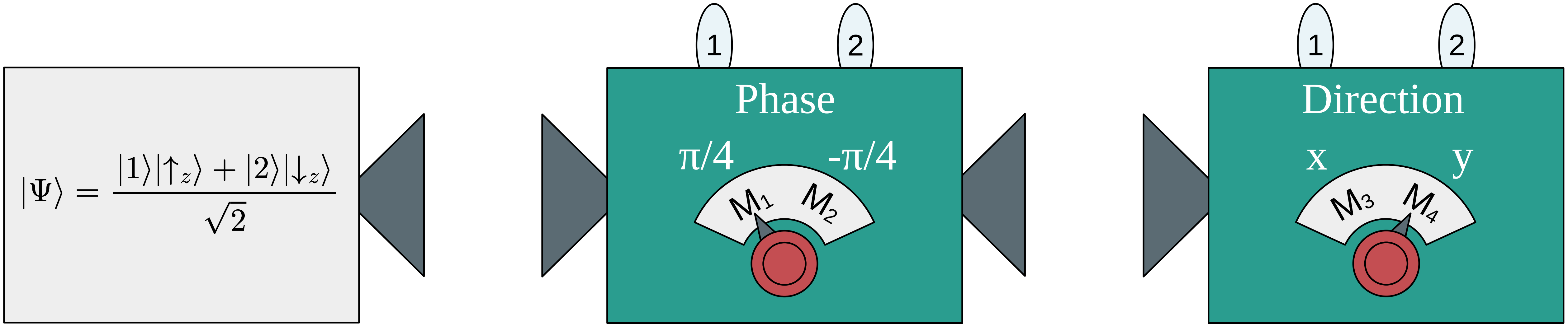
Path (M_1,M_2)

Spin (M_3,M_4)

Measurements:

$$\left[\hat{\Pi}_{k|M_i}, \hat{\Pi}_{l|M_j}\right] \neq 0$$

$$\left[\hat{\Pi}_{k|M_i}, \hat{\Pi}_{l|M_j}\right] = 0$$

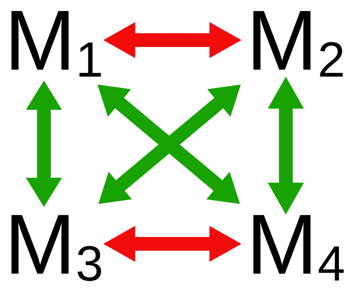


Preparation (P)

Path (M_1,M_2)

Spin (M_3,M_4)

Measurements:



Disturbing

Non-disturbing

$$\left[\hat{\Pi}_{k|M_i}, \hat{\Pi}_{l|M_j}\right] \neq 0$$

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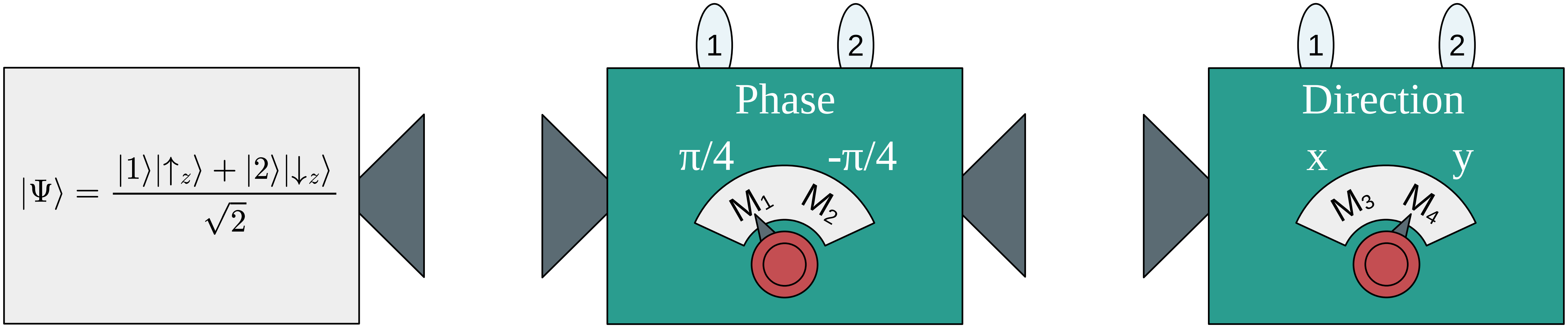

Preparation (P)

Path (M_1,M_2)

Spin (M_3,M_4)

Noncontextuality

Kochen, S. and Specker, E.P. "The Problem of Hidden Variables in Quantum Mechanics." Journal of Mathematics and Mechanics, vol. 17, no. 1, 1967, pp. 59–87. JSTOR

Neutron interferometry test of nonclassicality

Outcome probabilities:

$$\mathrm{Prob}(k, l | P, M_i, M_j) = \langle \Psi | \hat{\Pi}_{k|M_i} \hat{\Pi}_{l|M_j} | \Psi \rangle$$

$$\stackrel{?}{=} \sum_{\lambda} \mu(\lambda | P) \xi(k | M_i, \lambda) \xi(l | M_j, \lambda)$$

Neutron interferometry test of nonclassicality

Outcomes correlations:

$$E(M_i, M_j) = \mathrm{Prob}(k, k|P, M_i, M_j) - \mathrm{Prob}(k, l \neq k|P, M_i, M_j)$$

Clauser–Horne–Shimony–Holt (CHSH) inequality:

$$E(M_1, M_3) - E(M_1, M_4) + E(M_2, M_3) + E(M_2, M_4) \leq 2$$

J.F. Clauser; M.A. Horne; A. Shimony; R.A. Holt (1969), "Proposed experiment to test local hidden-variable theories", Phys. Rev. Lett., 23 (15): 880–4,

Neutron interferometry test of nonclassicality

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Eperimental results:

$$2.051 \pm 0.0019$$

Hasegawa, Yuji, et al. “Violation of a bell-like inequality in single-neutron interferometry.” Nature 425, 45–48 (2003).

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Hasegawa, Yuji, et al. “Violation of a bell-like inequality in single-neutron interferometry.” Nature 425, 45–48 (2003).

$$2.365 \pm 0.0013$$

Geppert, Hermann, et al. "Improvement of the polarized neutron interferometer setup demonstrating violation of a Bell-like inequality." Nuclear Instruments and Methods in Physics Research Section A 763 (2014): 417-423.

Neutron interferometry test of nonclassicality

Is quantum mechanics secretly classical?

...NO!

Neutron interferometry test of nonclassicality

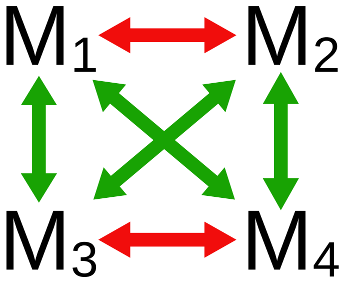
Is quantum mechanics secretly classical?

...NO!*

*according to our definition of non-disturbance (commutativity)

Bell test of nonclassicality

Measurements:

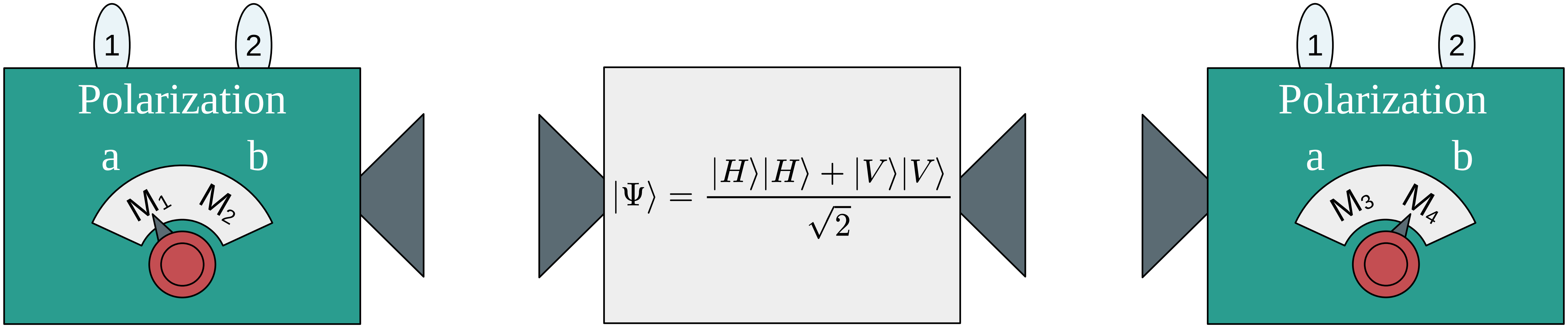


Disturbing

Non-disturbing

$$\left[\hat{\Pi}_{k|M_i},\hat{\Pi}_{l|M_j}\right]\neq 0$$

$$\left[\hat{\Pi}_{k|M_i},\hat{\Pi}_{l|M_j}\right]= 0$$



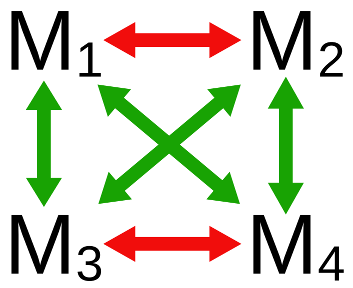
Particle 1 (M_1,M_2)

Preparation (P)

Particle 2 (M_3,M_4)

Bell test of nonclassicality

Measurements:

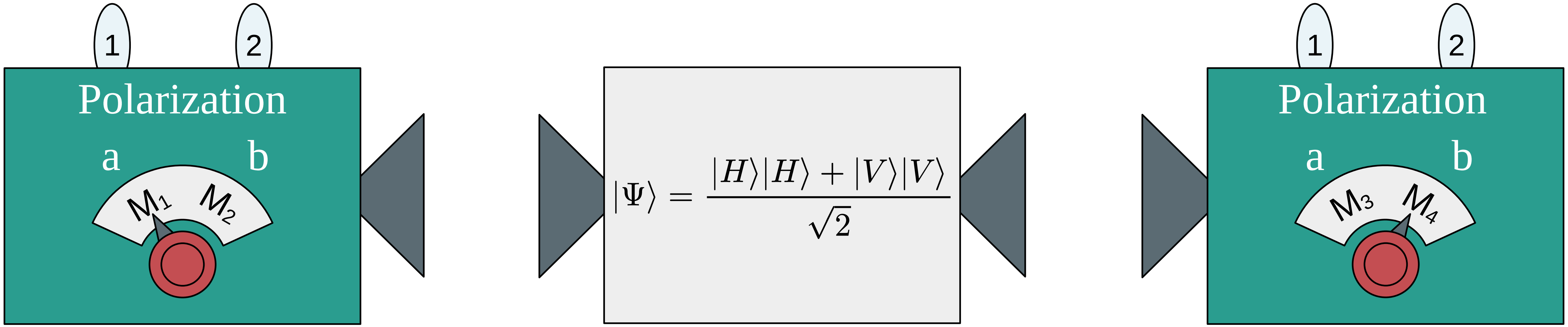


Disturbing

Non-disturbing

$$\left[\hat{\Pi}_{k|M_i}, \hat{\Pi}_{l|M_j}\right] \neq 0$$

Separated in space!



Particle 1 (M_1,M_2)

Preparation (P)

Particle 2 (M_3,M_4)

Locality

Bell, John S. "On the einstein podolsky rosen paradox." Physics Physique Fizika 1.3 (1964): 195.

Bell test of nonclassicality

Experimental result:

$$2.697 \pm 0.015 > 2$$

Aspect, Alain, et al. "Experimental realization of Einstein-Podolsky-Rosen-Bohm Gedankenexperiment: A new violation of Bell's inequalities." Physical review letters 49.2 (1982): 91.

Quantum mechanics seems to be nonclassical independently of the quantum model

Questions?

Break

Is quantum mechanics classical?

Outcome probabilities:

$$\mathrm{Prob}(k|P,M) = \langle\Psi|\hat{\Pi}_{k|M}|\Psi\rangle \overset{?}{=} \sum_\lambda \mu(\lambda|P)\xi(k|M,\lambda)$$

Quasiprobability representations and weak values

Quasiprobability representations:

$$\mathrm{Prob}(k|P,M) = \langle\Psi|\hat{\Pi}_{k|M}|\Psi\rangle = \sum_{\lambda} \tilde{\mu}(\lambda|P)\tilde{\xi}(k|\lambda,M)$$

Quasiprobability representations:

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Anomalous features:

$$\tilde{\mu}(\lambda|\Psi), \tilde{\xi}(k|\lambda,\hat{\Pi}_{k|M}) < 0$$

$$\tilde{\xi}(k|\lambda,\hat{\Pi}_{k|M}) > 1$$

$$\tilde{\mu}(\lambda|\Psi), \tilde{\xi}(k|\lambda,\hat{\Pi}_{k|M}) \in \mathbb{C}$$

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Ferrie, Christopher. "Quasi-probability representations of quantum theory with applications to quantum information science." Reports on Progress in Physics 74.11 (2011): 116001.
Spekkens, Robert W. "Negativity and contextuality are equivalent notions of nonclassicality." Physical review letters 101.2 (2008): 020401.

Quasiprobability representations:

Quasiprobability distribution

$$\mathrm{Prob}(k|P,M) = \langle\Psi|\hat{\Pi}_{k|M}|\Psi\rangle = \sum_{\lambda} \tilde{\mu}(\lambda|P)\tilde{\xi}(k|\lambda,M)$$

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Quasiprobability representations

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Quasiprobability distribution

$$\mathrm{Prob}(k|P,M) = \langle\Psi|\hat{\Pi}_{k|M}|\Psi\rangle = \sum_{\lambda} \tilde{\mu}(\lambda|P)\tilde{\xi}(k|\lambda,M)$$

Symbol

Anomalous features:

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Quasiprobability representations

Two orthonormal basis: $\{|a_i\rangle\}, \{|b_j\rangle\}$

Kirkwood-Dirac quasiprobability representation:

$$\mathrm{Prob}(k|P, M) = \langle\Psi|\hat{\Pi}_{k|M}|\Psi\rangle = \sum_{a_i, b_j} \langle\Psi|b_j\rangle\langle b_j|a_i\rangle\langle a_i|\Psi\rangle \, \frac{\langle b_j|\hat{\Pi}_{k|M}|a_i\rangle}{\langle b_j|a_i\rangle}$$

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Weak value

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Weak value

..and we can measure it!

Quasiprobability representations

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$$= \sum_{a_i,b_j} |\langle b_j|\Psi\rangle|^2 \, \frac{\langle b_j|a_i\rangle\langle a_i|\Psi\rangle}{\langle b_j|\Psi\rangle} \, \frac{\langle b_j|\hat{\Pi}_{k|M}|a_i\rangle}{\langle b_j|a_i\rangle}$$

Weak values!

..and we can measure them!

A standard (Von Neumann) measurement

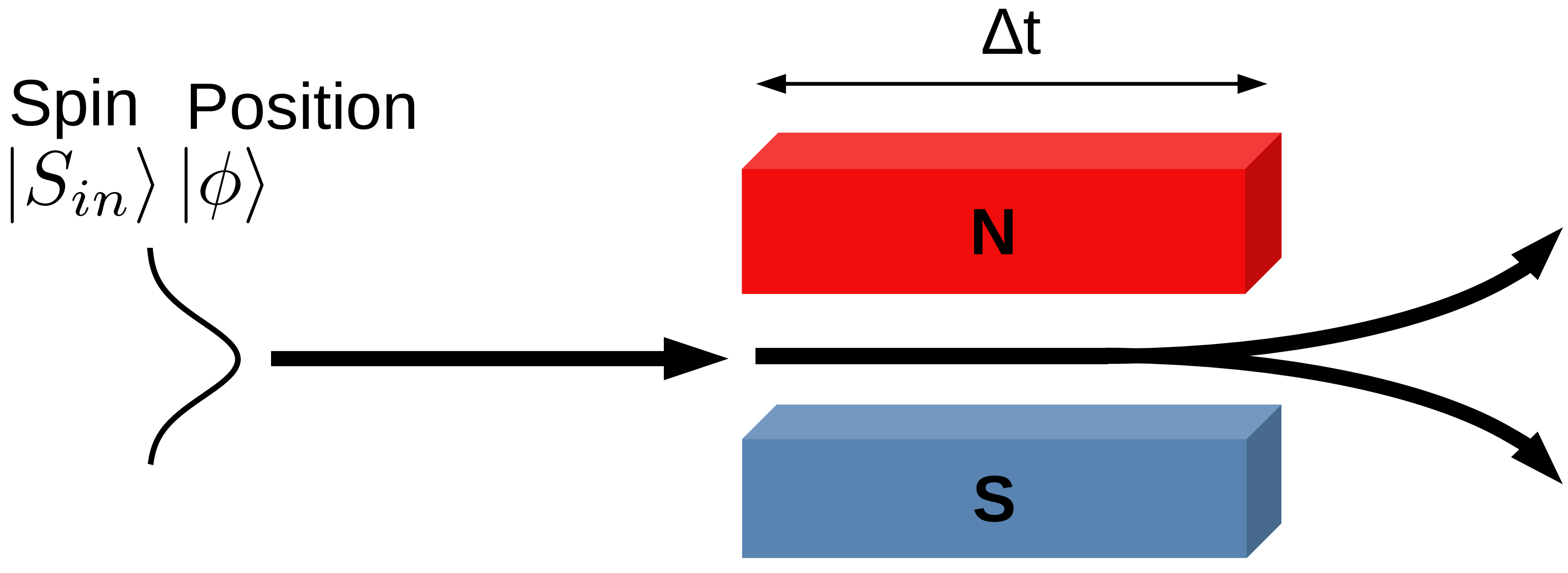
Stern-Gerlach

Spin Position

$|S_{in}\rangle\,|\phi\rangle$

A standard (Von Neumann) measurement

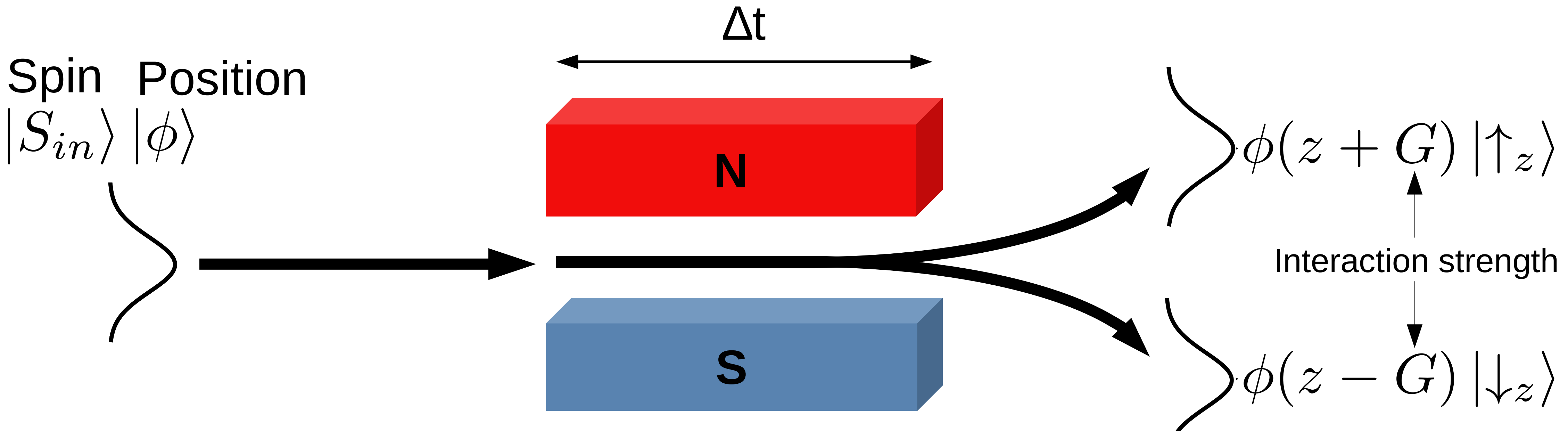
Stern-Gerlach



$$\hat{H}_I(t) \propto \hat{\sigma}_z \, \hat{p}_z$$

A standard (Von Neumann) measurement

Stern-Gerlach



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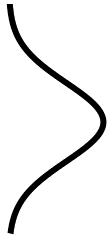
A weak measurement

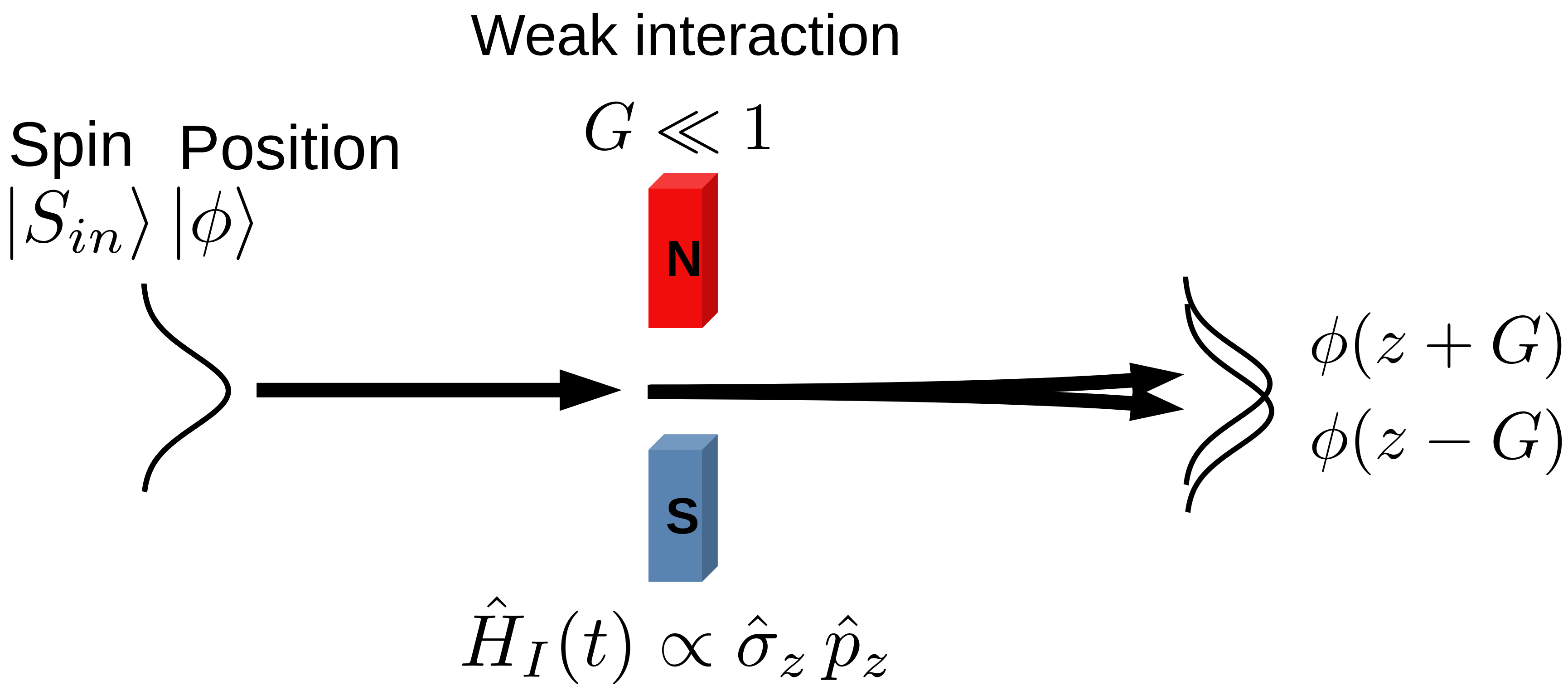
Spin Position

$|S_{in}\rangle \, |\phi\rangle$



Aharonov, Yakir, et al. "How the result of a measurement of a component of the spin of a spin-1/2 particle can turn out to be 100." Physical review letters 60.14 (1988): 1351.

A weak measurement



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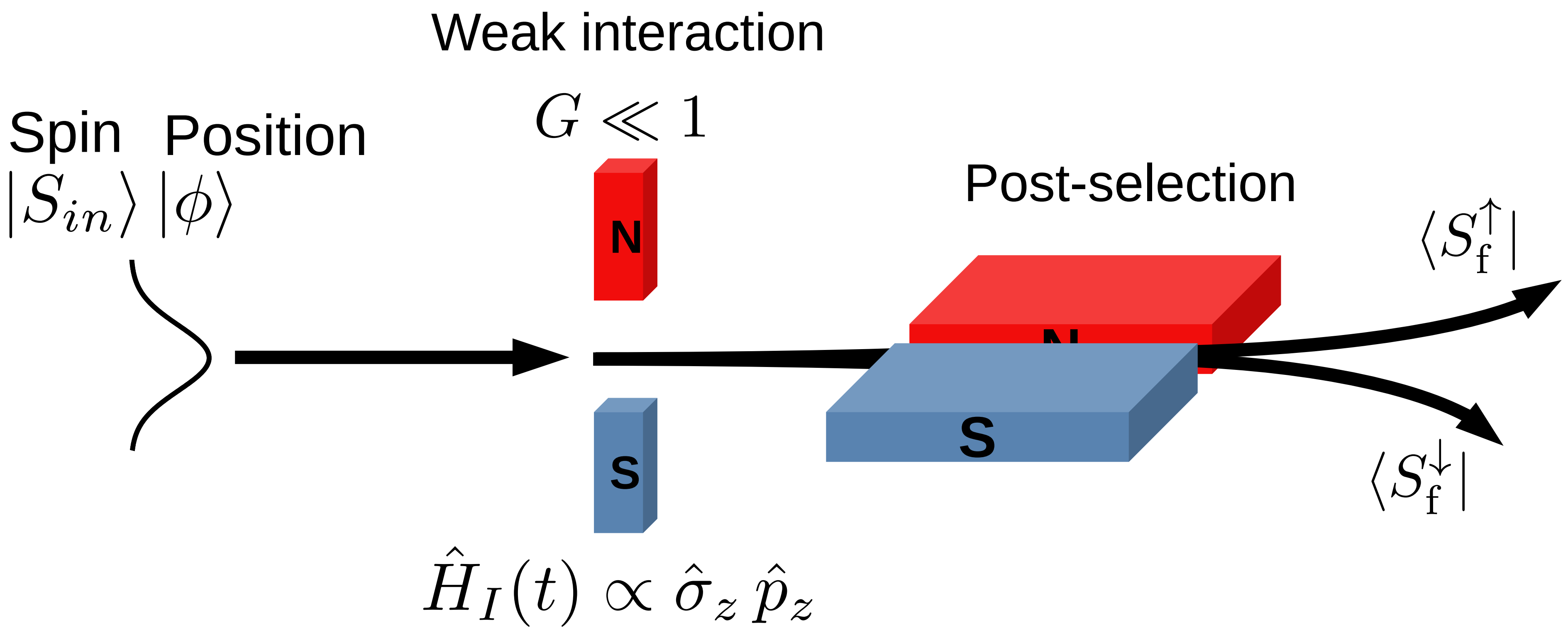

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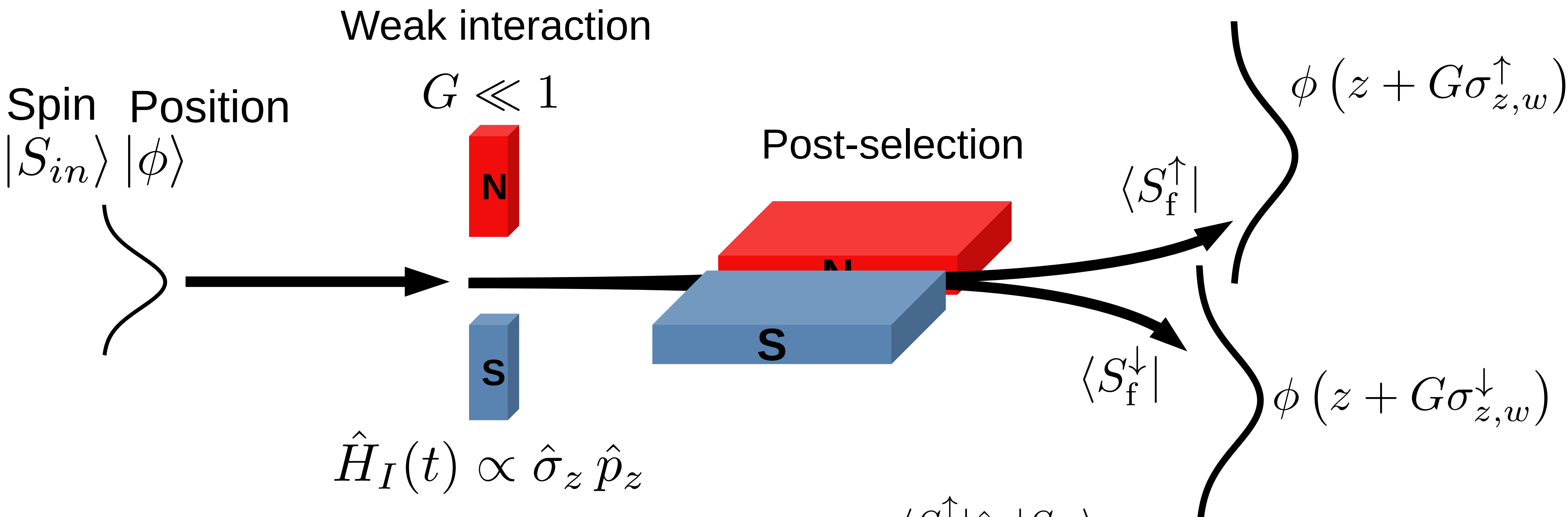
A weak measurement



Spin weak value: $\sigma_{z,w}^{\updownarrow} = \dfrac{\langle S_{\mathrm{f}}^{\updownarrow}|\hat{\sigma}_z|S_{\mathrm{in}}\rangle}{\langle S_{\mathrm{f}}^{\updownarrow}|S_{\mathrm{in}}\rangle}$

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Weak values extraction in neutron interferometry

Meter state and weak interaction:

- *Lemmel, Hartmut, et al. "Quantifying the presence of a neutron in the paths of an interferometer." Physical Review Research 4.2 (2022): 023075.*
- *Danner, Armin, et al. "Simultaneous path weak-measurements in neutron interferometry." Scientific Reports 14.1 (2024): 25994.*
- *Dvorak, Andreas, et al. "Tight qubit uncertainty relations studied through weak values in neutron interferometry." Physical Review Research 7.4 (2025): 043334.*

Meter state and no weak interaction:

- *Denkmayr, Tobias, et al. "Weak values from strong interactions in neutron interferometry." Physica B: Condensed Matter 551 (2018): 339-346.*

No meter state and weak interaction:

- *Masiello, Ismaele V., et al. "Simultaneous determination of two path weak-values with time-dependent phase manipulation in neutron interferometry." New Journal of Physics 27.2 (2025): 023017.*

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Weak values extraction in neutron interferometry

Meter states or weak interactions increase the experimental complexity!

The manipulation of additional degrees of freedom increases the amount of required experimental resources (additional instrumentation and procedures), while weak interactions typically necessitate longer measurement times due to the limited information gained per detection event.

Weak values extraction in neutron interferometry

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No meter state and no weak interaction:

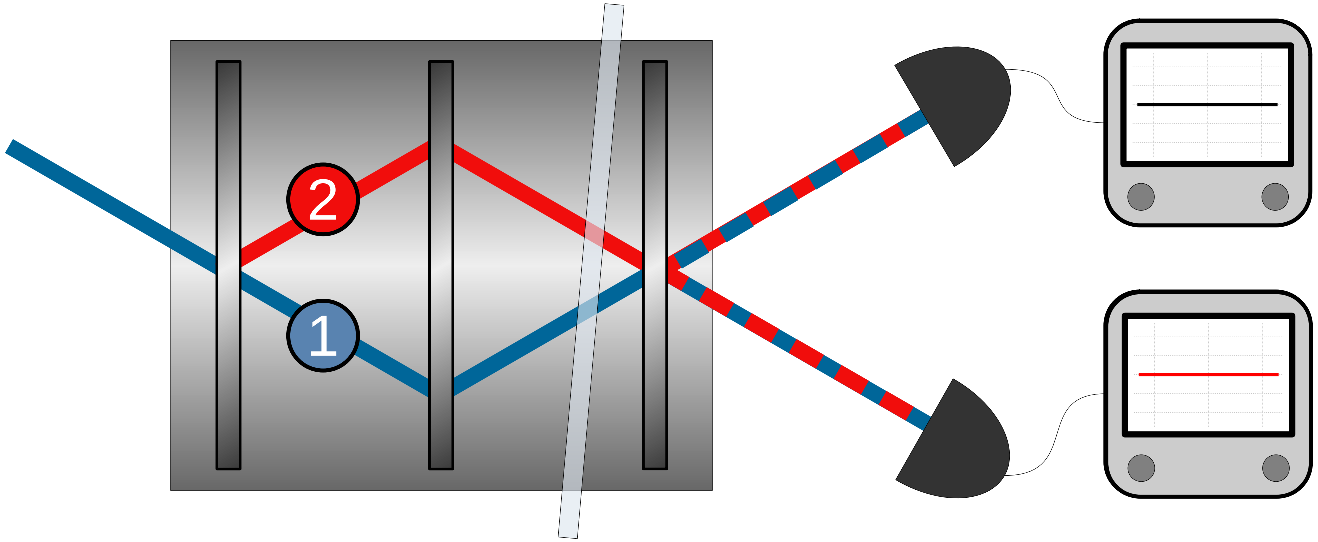
- *Masiello, Ismaele V., et al. "Anomalous weak values in a generalized Mach-Zehnder interferometer extracted directly from intensity measurements." in review at npj Quantum Information.*

Unexpectedly running out of time :(

Brief experimental results

Weak values from output intensities

Neutron Mach-Zehnder interferometer:



Initial state:

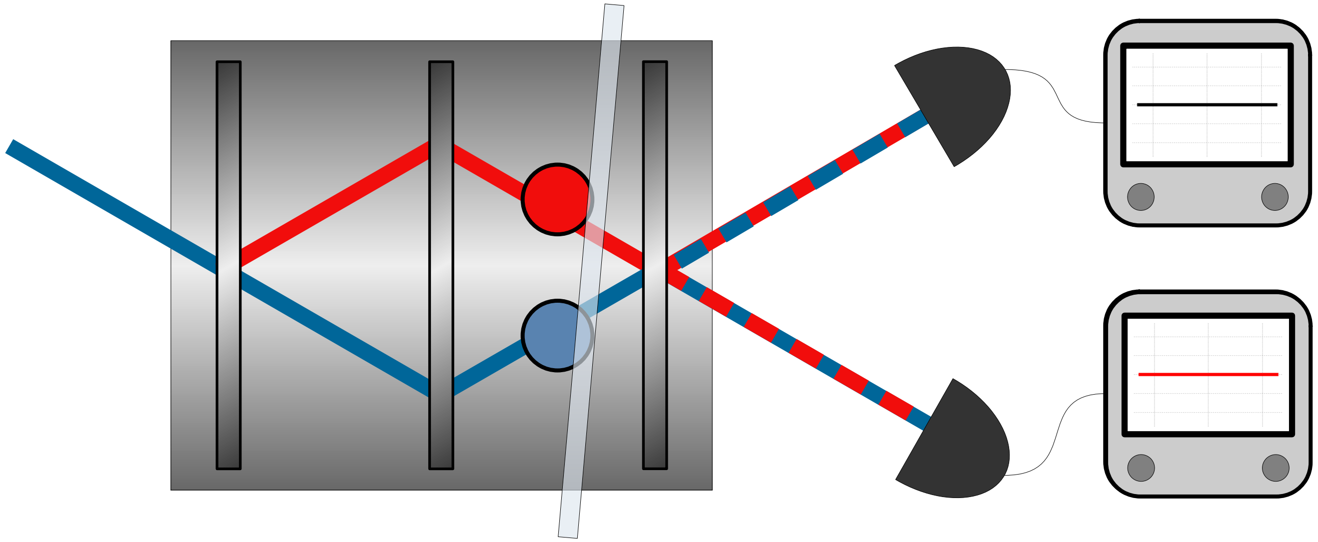
$$|\psi_{\mathrm{in}}\rangle = \cos\left(\frac{\theta}{2}\right)|1\rangle + \mathrm{e}^{\mathrm{i}\phi}\sin\left(\frac{\theta}{2}\right)|2\rangle$$

Path observable:

$$\hat{\Pi}_1 = |1\rangle\langle 1| \qquad \hat{\Pi}_2 = |2\rangle\langle 2|$$

Weak values from output intensities

Generalized neutron Mach-Zehnder interferometer:



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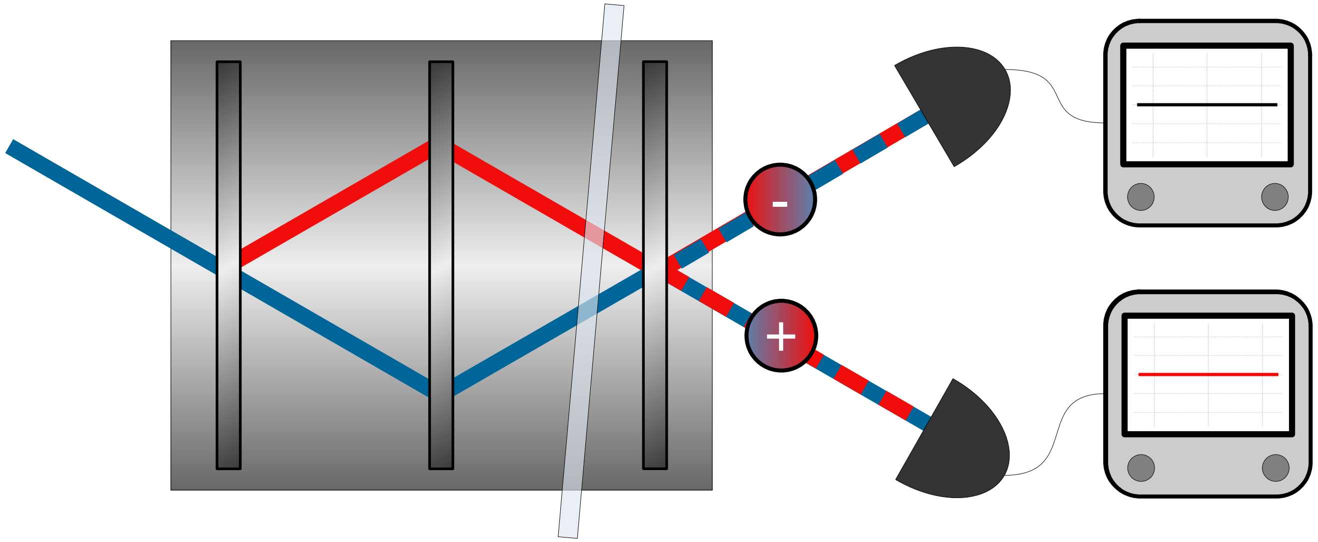
Phase shift:

$$|\psi_{\mathrm{in}}\rangle \rightarrow \mathrm{e}^{-\mathrm{i}\delta\hat{\Pi}_1}|\psi_{\mathrm{in}}\rangle$$

Weak values from output intensities

Generalized neutron Mach-Zehnder interferometer:



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Phase shift:

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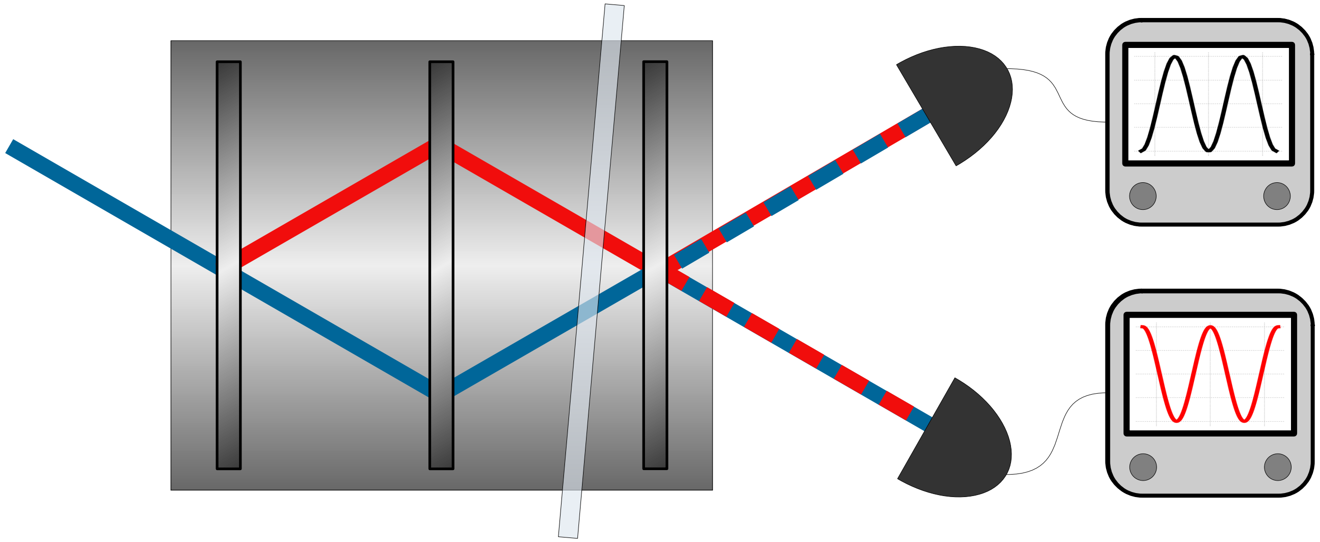
Final states:

$$|\psi_+\rangle = \frac{|1\rangle + |2\rangle}{\sqrt{2}} \quad |\psi_-\rangle = \frac{|1\rangle - |2\rangle}{\sqrt{2}}$$

Weak values from output intensities

Generalized neutron Mach-Zehnder interferometer:



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Final states:

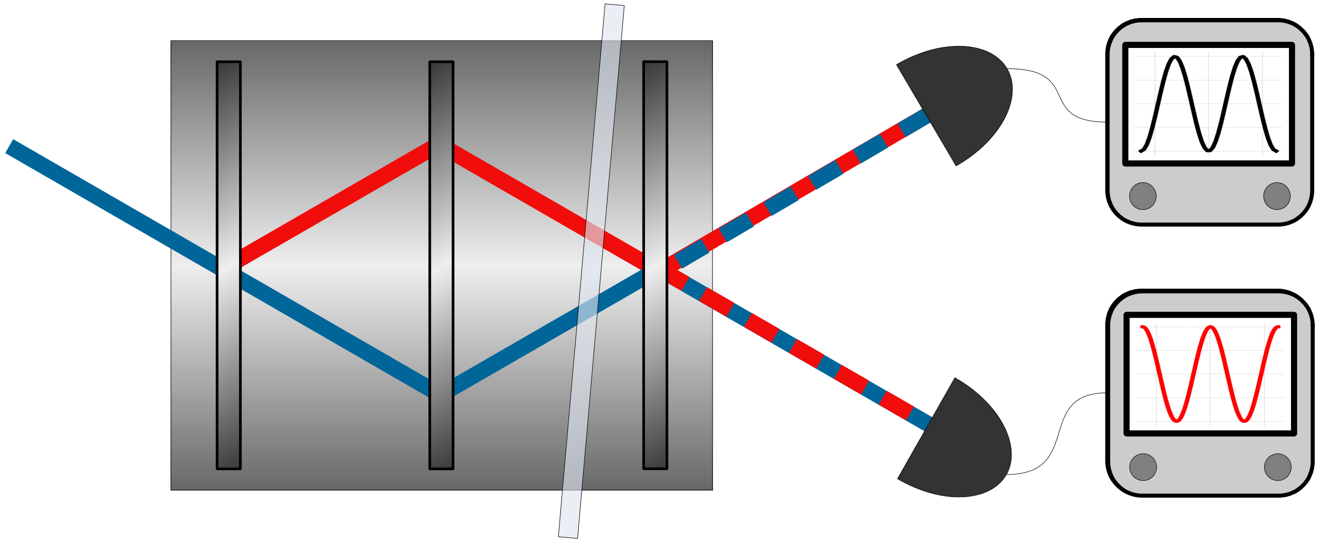
$$|\psi_+\rangle = \frac{|1\rangle + |2\rangle}{\sqrt{2}} \quad |\psi_-\rangle = \frac{|1\rangle - |2\rangle}{\sqrt{2}}$$

Measured intensity:

$$I_\pm(\delta) = A\left[\frac{1}{2} \pm \frac{1}{2}\sin\theta\cos(\phi+\delta)\right]$$

Weak values from output intensities

Generalized neutron Mach-Zehnder interferometer:



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Final states:

$$|\psi_+\rangle = \frac{|1\rangle + |2\rangle}{\sqrt{2}} \quad |\psi_-\rangle = \frac{|1\rangle - |2\rangle}{\sqrt{2}}$$

Path weak-values:

$$w_{\pm,j} = \frac{\langle\psi_\pm|\hat{\Pi}_j|\psi_{\rm in}\rangle}{\langle\psi_\pm|\psi_{\rm in}\rangle} = w^{\rm R}_{\pm,j} + i\, w^{\rm I}_{\pm,j}$$

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Weak values from output intensities

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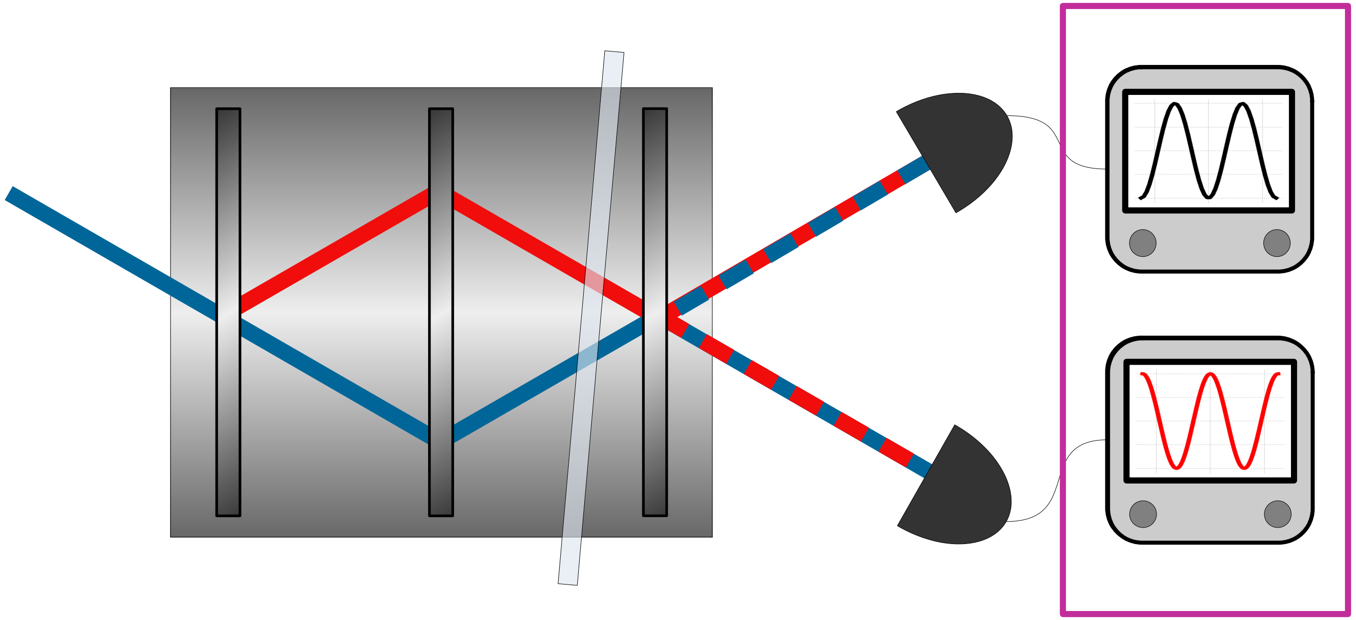


Weak values can be directly extracted from the output intensities!

Path weak-values:

$$w_{\pm,j} = \frac{\langle \psi_\pm | \hat{\Pi}_j | \psi_{\text{in}} \rangle}{\langle \psi_\pm | \psi_{\text{in}} \rangle} = w^{\text{R}}_{\pm,j} + \mathrm{i}\, w^{\text{I}}_{\pm,j}$$

Weak values from output intensities

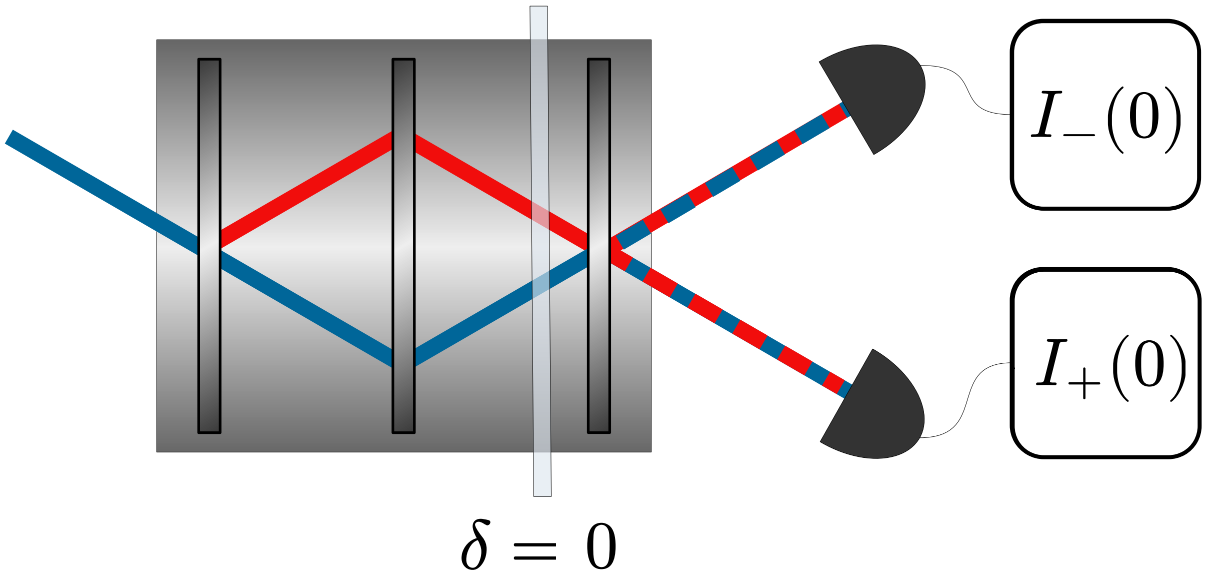
Measured intensity:

$$I_{\pm}(\delta) = A\,|\langle\psi_{\pm}|\psi_{\text{in}}\rangle|^2\,\left[1 + 2\left(|w_{\pm,1}|^2 - w^{\text{R}}_{\pm,1}\right)(1-\cos\delta) + 2\,w^{\text{I}}_{\pm,1}\,\sin\delta\right]$$

Amplitude squared   Real part   Imaginary part

Weak values from output intensities

Three measurements settings:



Weak values from output intensities

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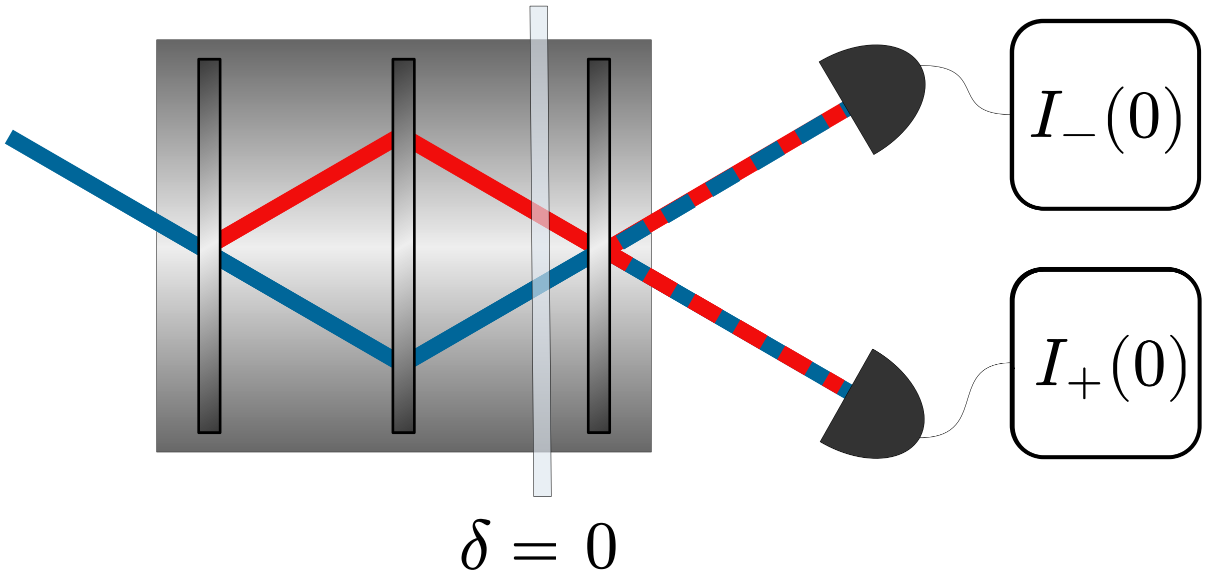
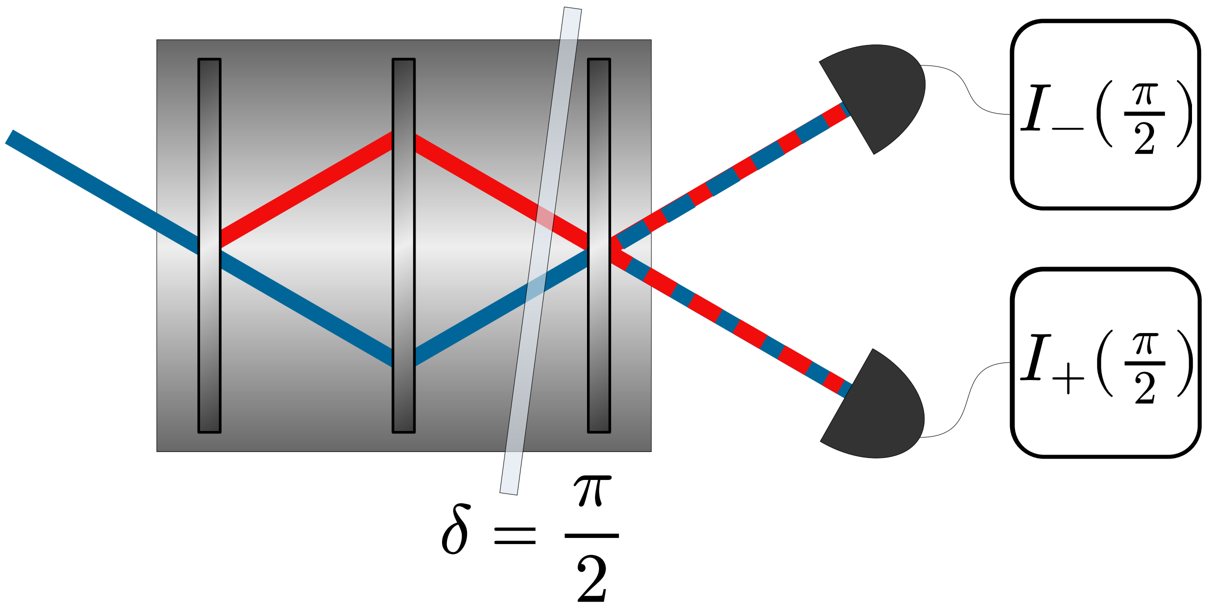
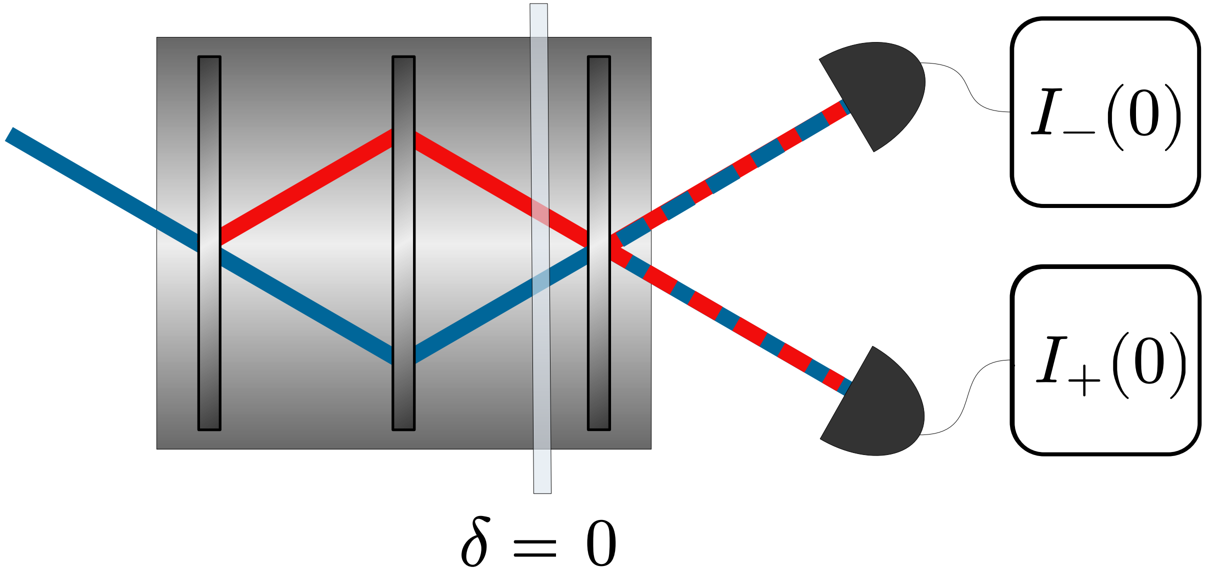
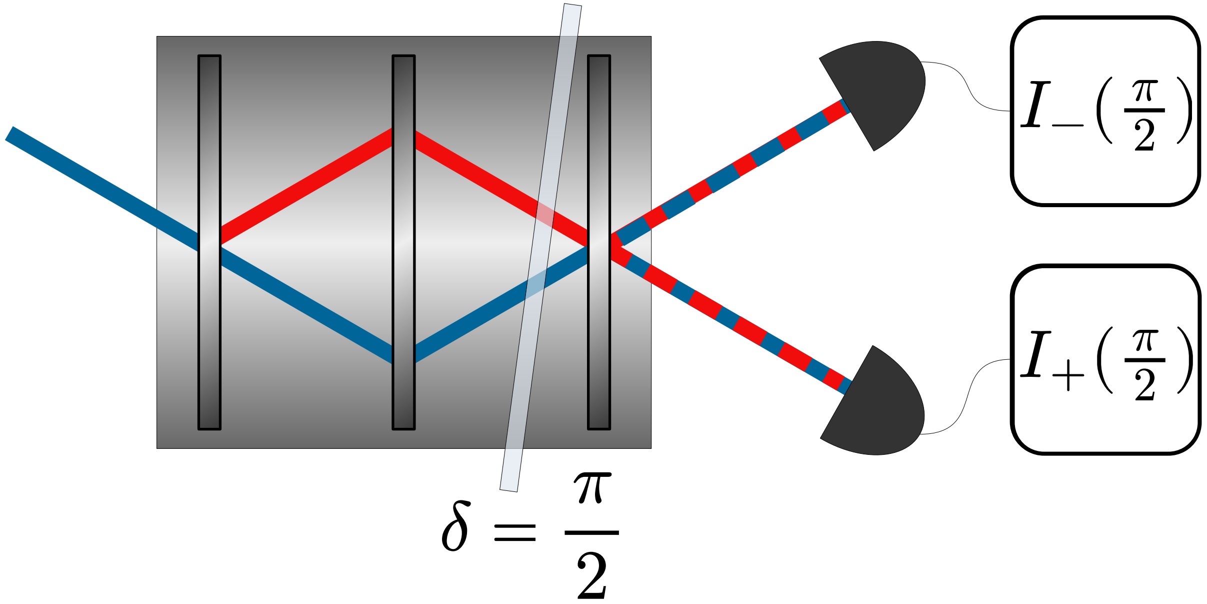


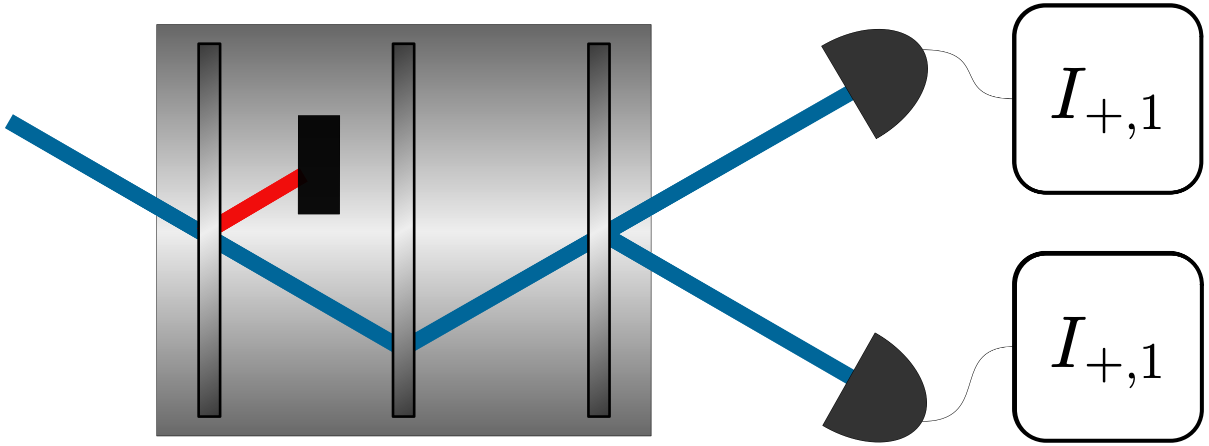


Weak values from output intensities

Three measurements settings:







Weak values from output intensities

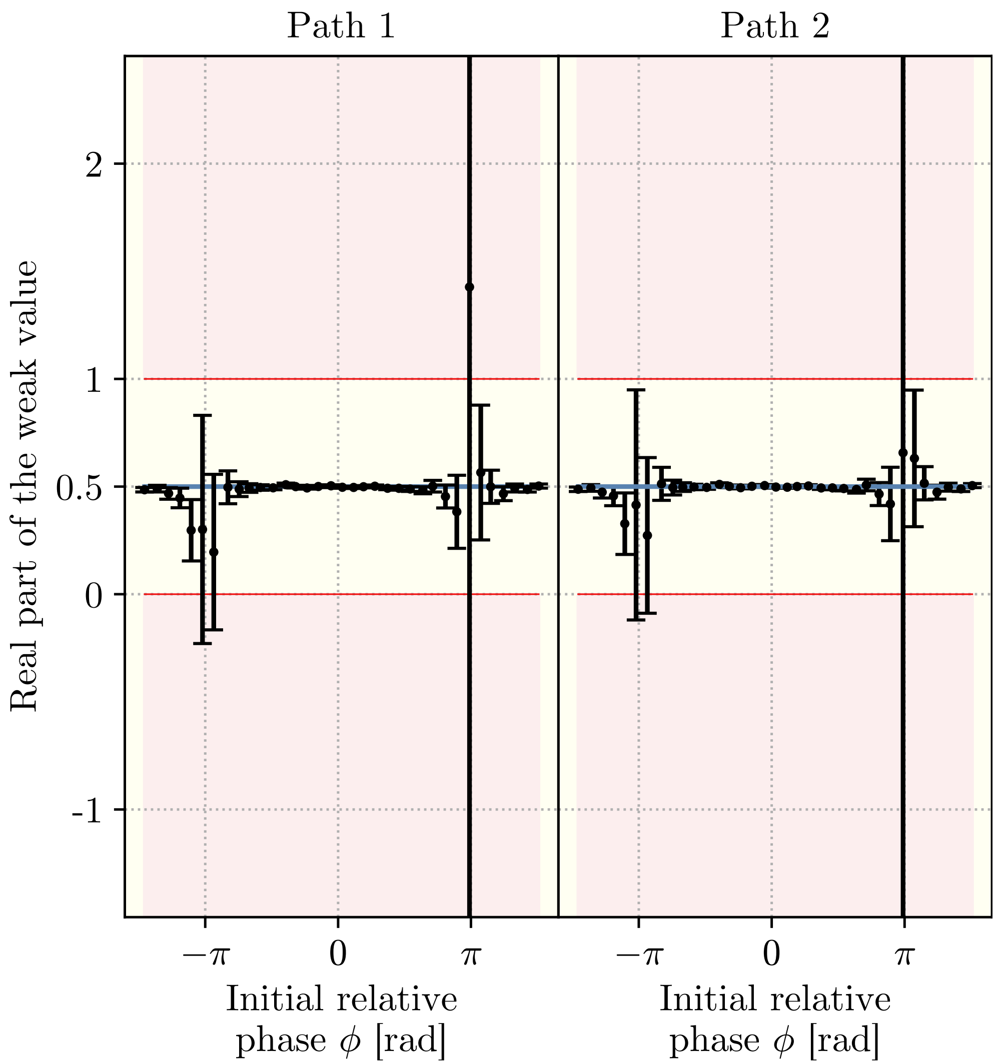
Weak value extraction:

$$w^{\mathrm{R}}_{+,1} = \frac{I_{+,1}}{I_+(0)} + \frac{I_+(0) - I_-(0)}{4I_+(0)}$$

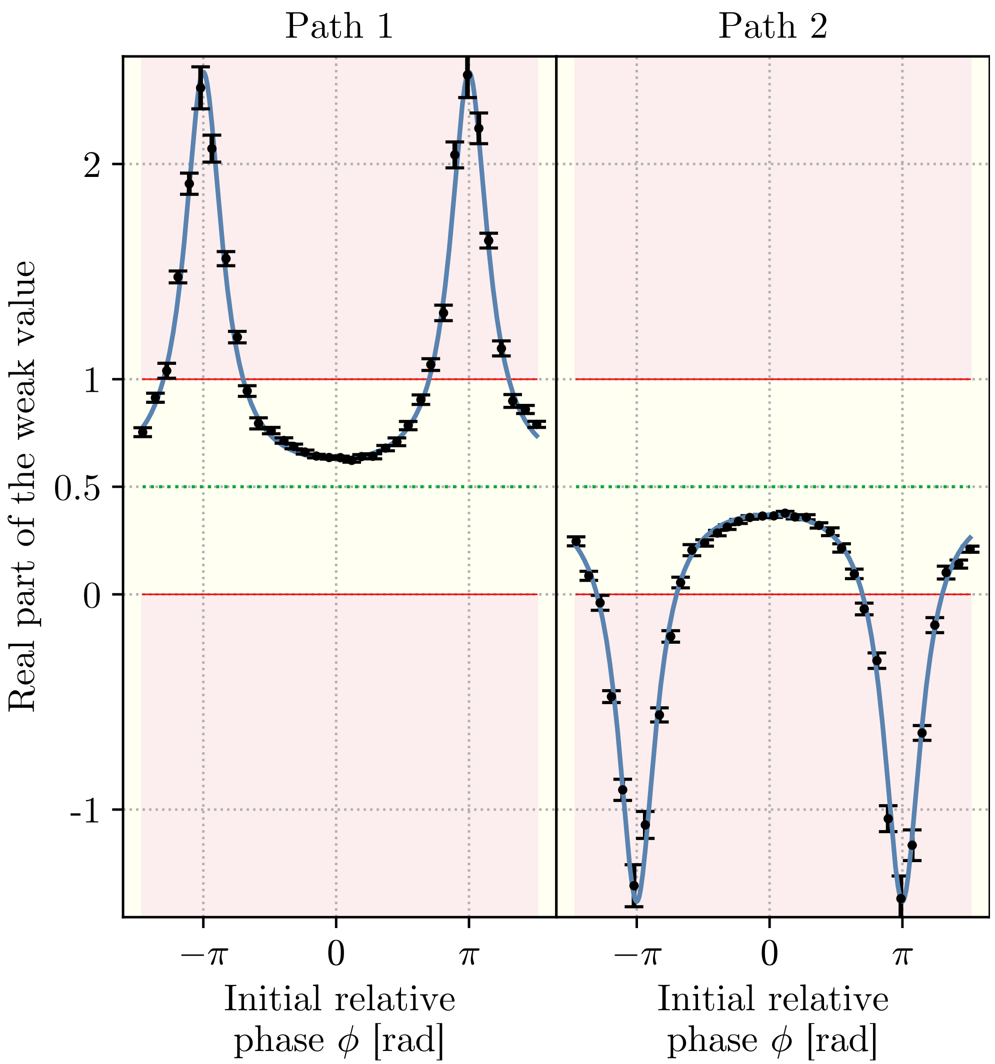
$$w^{\mathrm{I}}_{+,1} = \frac{I_+(\frac{\pi}{2}) - I_-(\frac{\pi}{2})}{4I_+(0)}$$

Weak values from output intensities

Balanced: $|\psi_{\mathrm{in}}\rangle \approx \dfrac{|1\rangle + \mathrm{e}^{\mathrm{i}\phi}|2\rangle}{\sqrt{2}}$

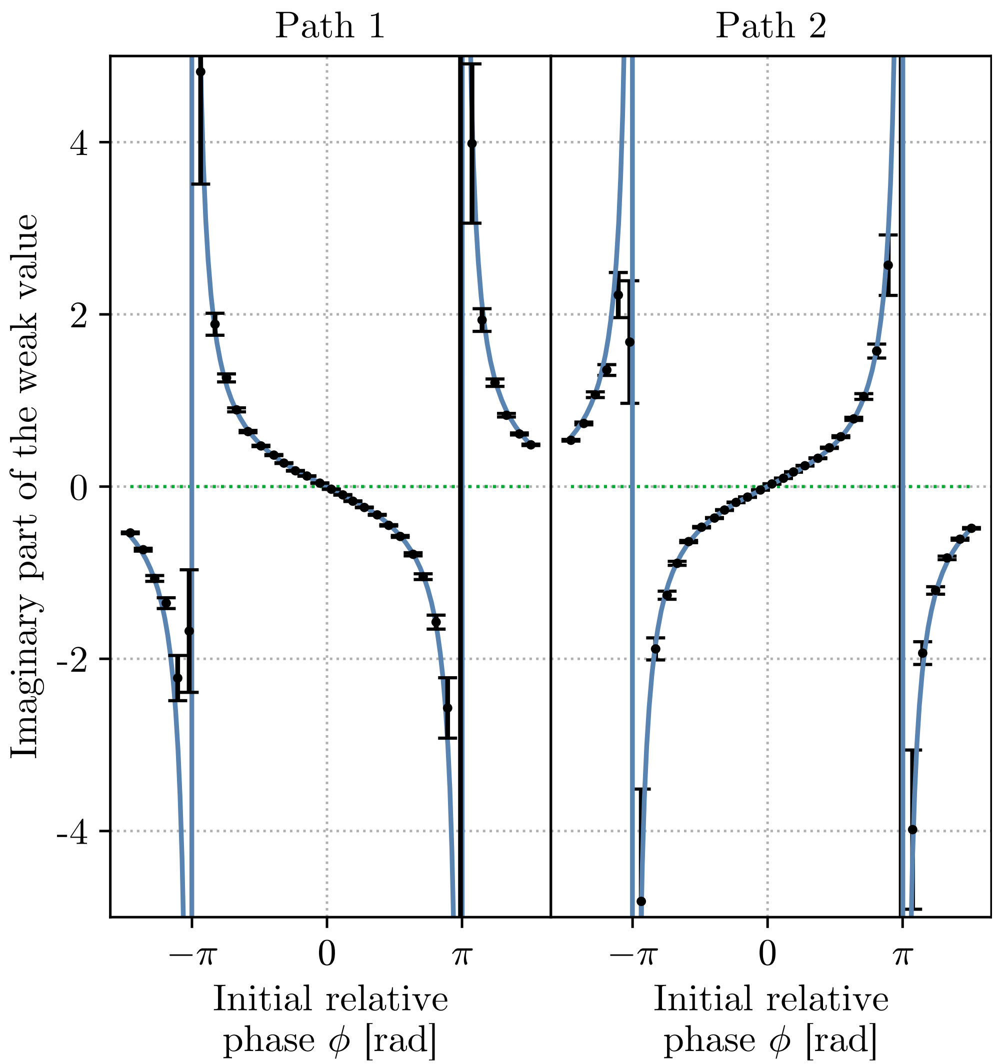


Unbalanced: $|\psi_{\mathrm{in}}\rangle \approx 0.86|1\rangle + 0.51\,\mathrm{e}^{\mathrm{i}\phi}|2\rangle$

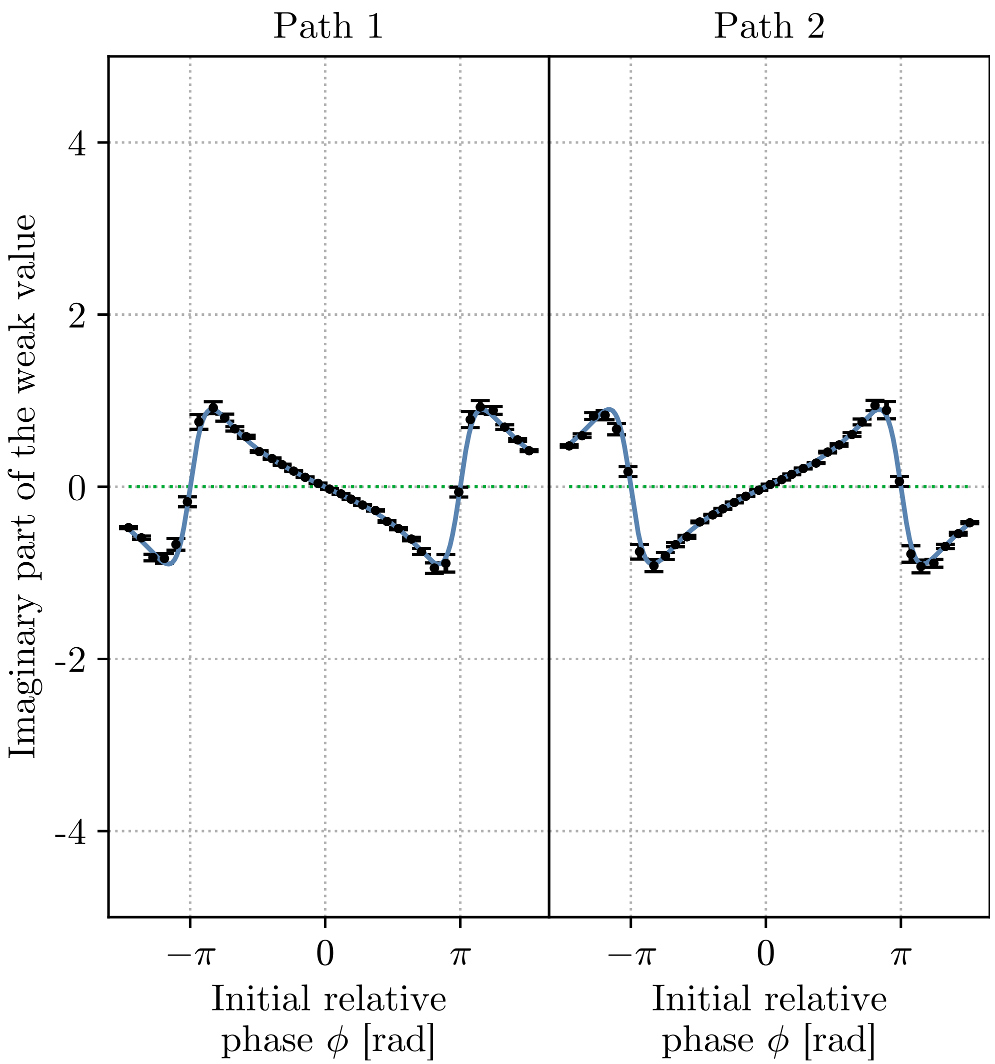


Weak values from output intensities

Balanced: $|\psi_{\rm in}\rangle \approx \dfrac{|1\rangle + \mathrm{e}^{\mathrm{i}\phi}|2\rangle}{\sqrt{2}}$



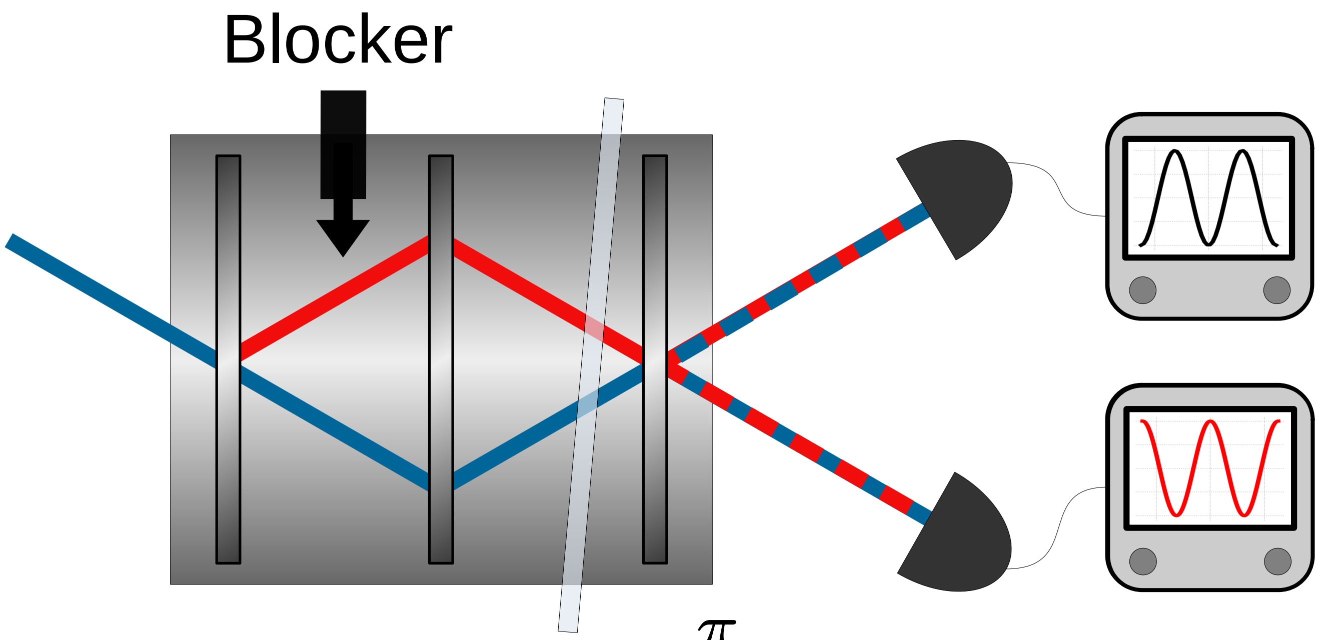
Unbalanced: $|\psi_{\rm in}\rangle \approx 0.86|1\rangle + 0.51\,\mathrm{e}^{\mathrm{i}\phi}|2\rangle$



Weak values from output intensities

Required setup:

- Interferometer
- Path blocker
- Phase shifter



$$\delta = 0, \frac{\pi}{2}$$

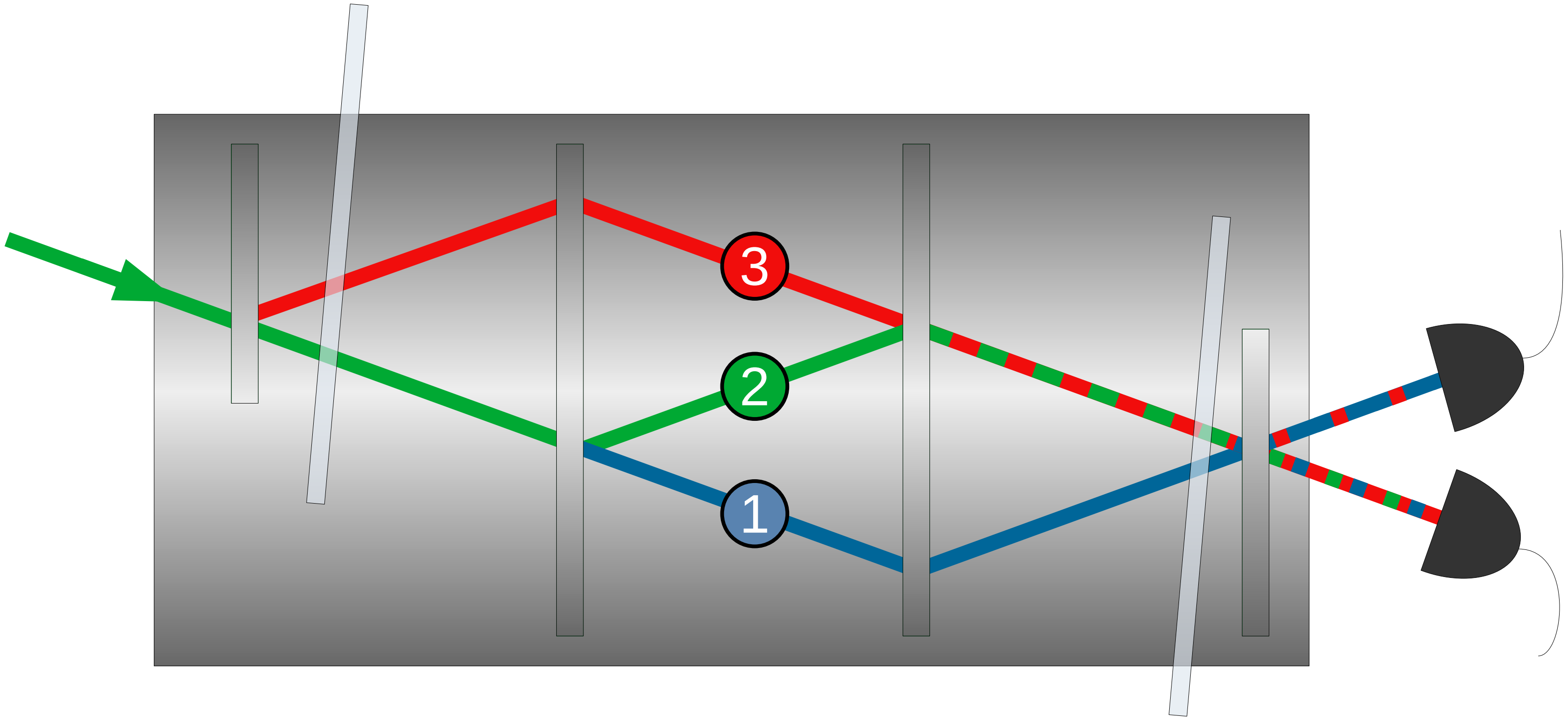
Measurements:

- No meter state and no weak interaction (shorter acquisition times with comparable or improved precision)
- 3 measurements needed to completely determine the path weak-values

Weak values from output intensities

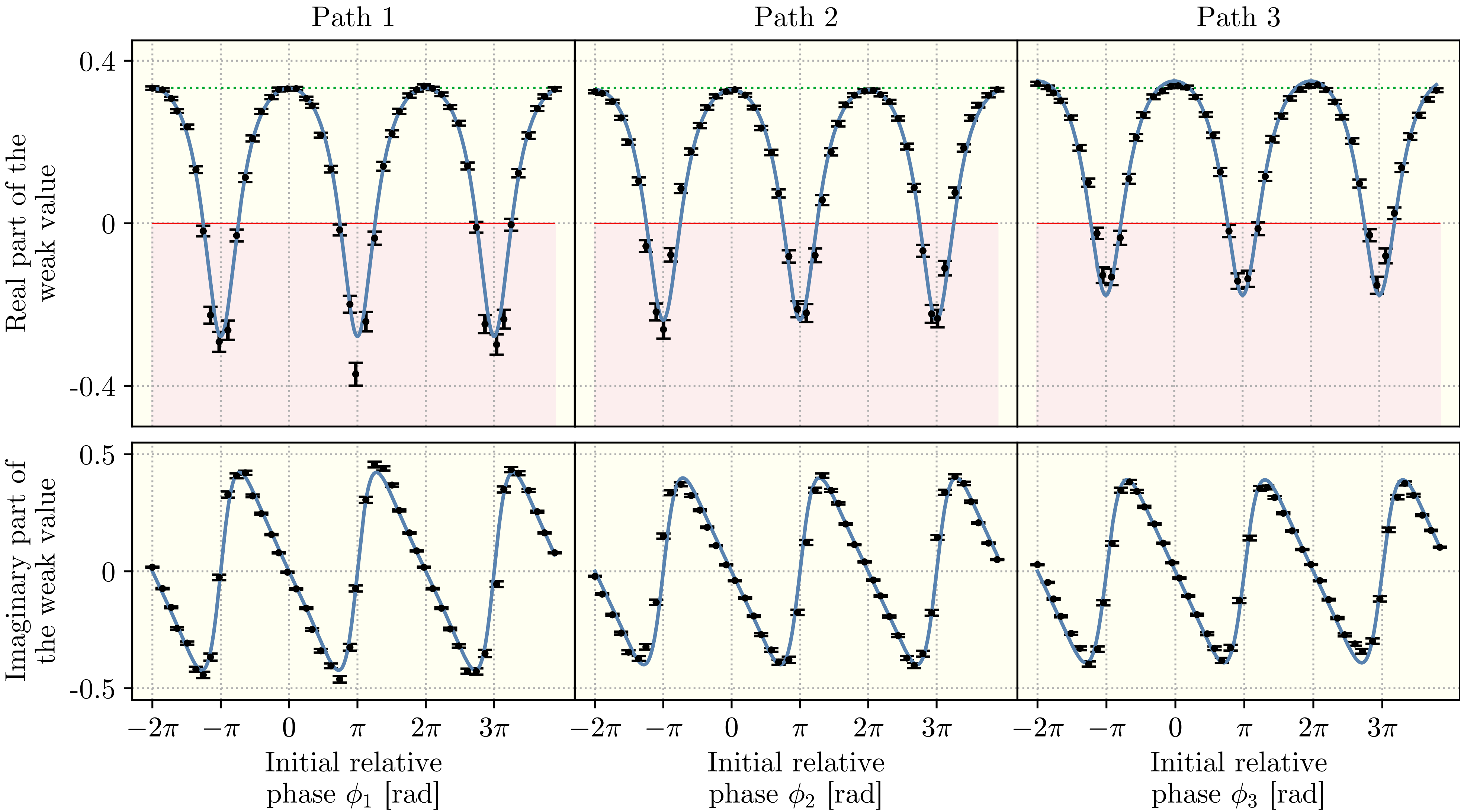
Bonus experimental result!

Three-path neutron interferometer:



Weak values from output intensities

Initial state: $|\psi_{\text{in}}\rangle \approx \dfrac{e^{i\phi_1}|1\rangle + e^{i\phi_2}|2\rangle + e^{i\phi_3}|3\rangle}{\sqrt{3}}$



Summary

- Quantum mechanics is nonclassical, but not always
- Weak values can be used to identify nonclassical phenomena
- Our group developed several methods to extract weak values in neutron interferometry
- In particular, one method that requires no weak interaction and no meter state was presented. The method can be applied any N-level quantum system in which alike measurements can be performed.

Neutron Interferometry Group

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Thank you for your attention!